FAN COILS
ACCESSORY ELECTRIC HEATERS

WIRING DIAGRAMS

FIG.	FIELD– INSTALLED HEATER MODEL	FB4C/ PF4MNP	FE4A/FE5A	FH4C	FV4C	FX4D	FY5B	FZ4A	PF4MA	PF4MB	LABEL
1	KFCEH0401N03A	18,24	x	001	x	19,25	18,24	24	18,19,24,25	x	340816–101
1	KFCEH0501N05A	18-60	002-006	001-002	002-006	19-61	18-60	24–61	18-61	19-61	340816–101
2	KFCEH0801N08A	18-60	002-006	001-003	002-006	19-61	18-60	24–61	18-61	19-61	340817–101
2	KFCEH0901N10A	18-60	002-006	001-004	002-006	19-61	18-60	24–61	18-61	19-61	340817–101
6	KFCEH1601315A	42-60	002-006	001-004	002-006	43-61	18-60	48–61	18-61	19-61	340819–101
7	KFCEH2001318A	42-60	003-006	001-004	002-006	43-61	42-60	48–61	42-61	37-61	340818–101
1	KFCEH2401C05A	18-60	002-006	001-002	002-006	19-61	18-60	24–61	18-61	19-61	340816–101
2	KFCEH2501C08A	18-60	002-006	001-003	002-006	19-61	18-60	24–61	18-61	19-61	340817–101
2	KFCEH2601C10A	18-60	002-006	001-004	002-006	19-61	18-60	24–61	18-61	19-61	340817–101
3	KFCEH2901N09A	36-60	002-006	003-004	002-006	37-61	36-60	36–61	36-60	31-61	340813–101
4	KFCEH3001F15A	24-60	002-006	001-004	002-006	25-61	24-60	24–61	24-61	19-61	340814–101
4	KFCEH3101C15A	24-60	002-006	001-004	002-006	25-61	24-60	24–61	24-61	19-61	340814–101
5	KFCEH3201F20A	30-60	002-006	002-004	002-006	31-61	30-60	36–61	30-61	19-61	340815–101
5	KFCEH3301C20A	30-60	002-006	002-004	002-006	31-61	30-60	36–61	30-61	19-61	340815–101
8,9	KFCEH3401F24A	48,60	004-006	003-004	005-006	49-61	48-60	48–61	48-61	49-61	340821–101, 340820–101
8,9	KFCEH3501F30A	48,60	004-006	003-004	005-006	49-61	48-60	48–61	48-61	49-61	340821–101 340820–101

FAN COIL WITH RBC X–13 MOTOR OR BROAD OCEAN DIGI MOTOR

FIG.	FACTORY-INSTALLED HEATER MODEL	FB4C	FX4D	FZ4A	LABEL
11	MKFCEH0501N05A	18,24	19,25	--	341018-101
10	MKFCEH0801N08A	18,30	31	--	341017-101
10	MKFCEH0901N10A	24-48	37-61	--	
12	MKFCEH3001F15A	36-60	61	--	341019-101
30	MKFCEH0801N08A	--	--	36,48	342416-101
	MKFCEH0901N10A	--	--		
31	MKFCEH3001F15A	--	--	60,61	342418-101
32	MKFCEH0501N05A	--	--	24	342417-101

FAN COIL WITH COOLING ONLY CONTROL

FIG.	MODEL	SIZE	LABEL
13	FV4C	002-006	326014-101
14	FE4A/FE5A	002-006	333107-101
15	FY5B/PF4MNA	18-60	328964-101
15	FH4C	001-004	328964-101
16	FB4C/FX4D/PF4MNP (RBC)	18-61	336228-101
16	PF4MNA/B	19,25,31,37,43,49,61	336228-101
17	FB4C/FX4D/PF4MNP (BOM)	18-61	337519-101
33	FZ4A	24-61	342415-101

ELECTRIC HEATERS

FIG.	HEATER MODEL	FF1E	LABEL
18	KFDEH0801D05A	18,24,30,36	341080-101
19	KFDEH0901D75A	18,24,30,36	341081-101
19	KFDEH1001D11A	18,24,30,36	341081-101
20	KFEEH0101D05A	19,25,31,37	341082-101
21	KFEEH0201D75A	19,25,31,37	341083-101
21	KFEEH0301D11A	19,25,31,37	341083-101

FIG.	HEATER MODEL	CONTROL TYPE	FFMA	LABEL
22	EHK2-05B	Sequencer - HS	18,24,30,36 Prior to serial number date code 1715V.	202070290385
22	EHK2-08B	Sequencer - HS	18,24,30,36 Prior to serial number date code 1715V.	202070290385
22	EHK2-10B	Sequencer - HS	18,24,30,36 Prior to serial number date code 1715V.	202070290385
23			18-37 Serial number date code 1715V and later.	2020702A717
24	EHK2-05B	Relay - HR	18-37	06-7094-02
24	EHK2-08B	Relay - HR	18-37	06-7094-02
24	EHK2-10B	Relay - HR	18-37	06-7094-02

FIG.	FFMA	LABEL
25	19,31	--
26	25,37	--

FIG.	FPM(A,B)N(C,U)	LABEL
27	ALL with Time Delay Relay	202070290388 Valid for models FPM(A,B)N(C,U)0**000AAAA
28	ALL with Time Delay Board	2020702A1716 Valid for models FPM(A,B)N(C,U)T00ACAA

FIG.	HEATER MODEL	FPMAN(C,U)	FPMBN(C,U)	LABEL
29	EHK3-05B	18-36	18-30	06-7094-03
29	EHK3-08B	18-36	18-30	06-7094-03
29	EHK3-10B	18-36	18-30	06-7094-03

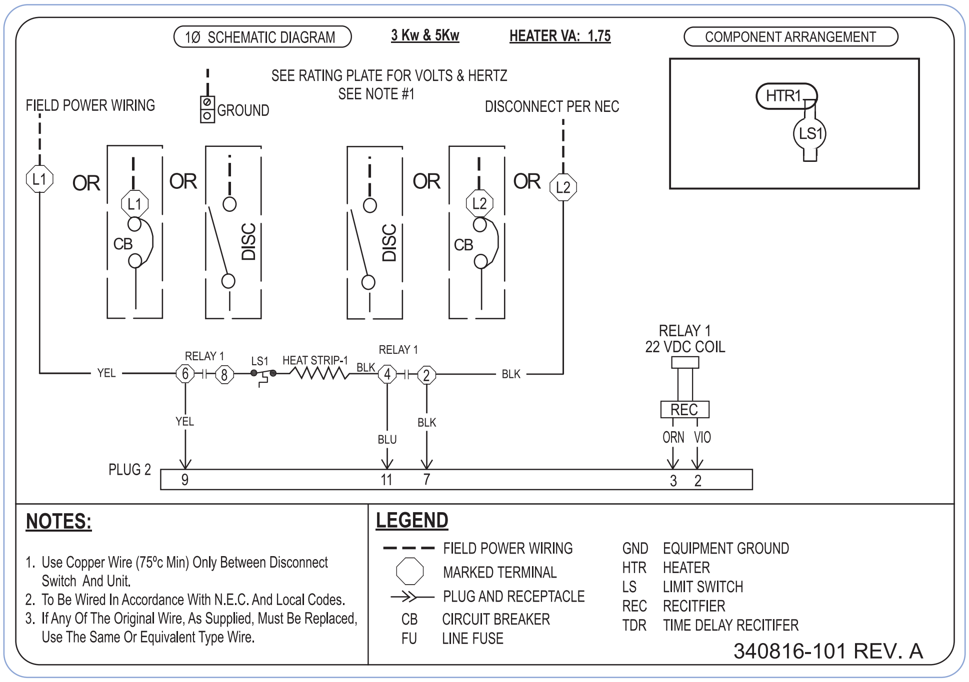
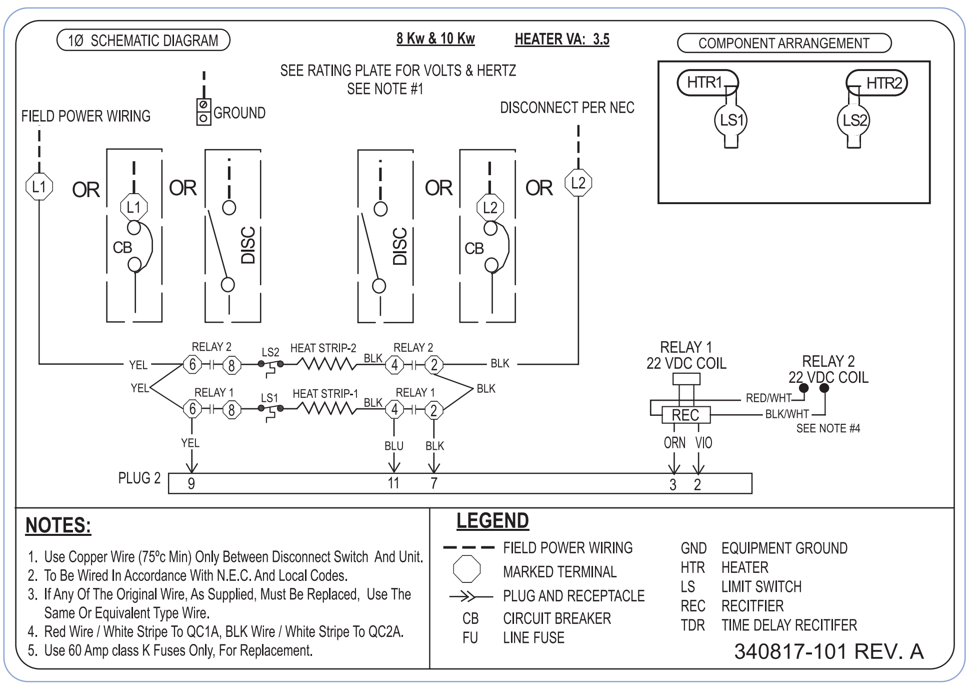


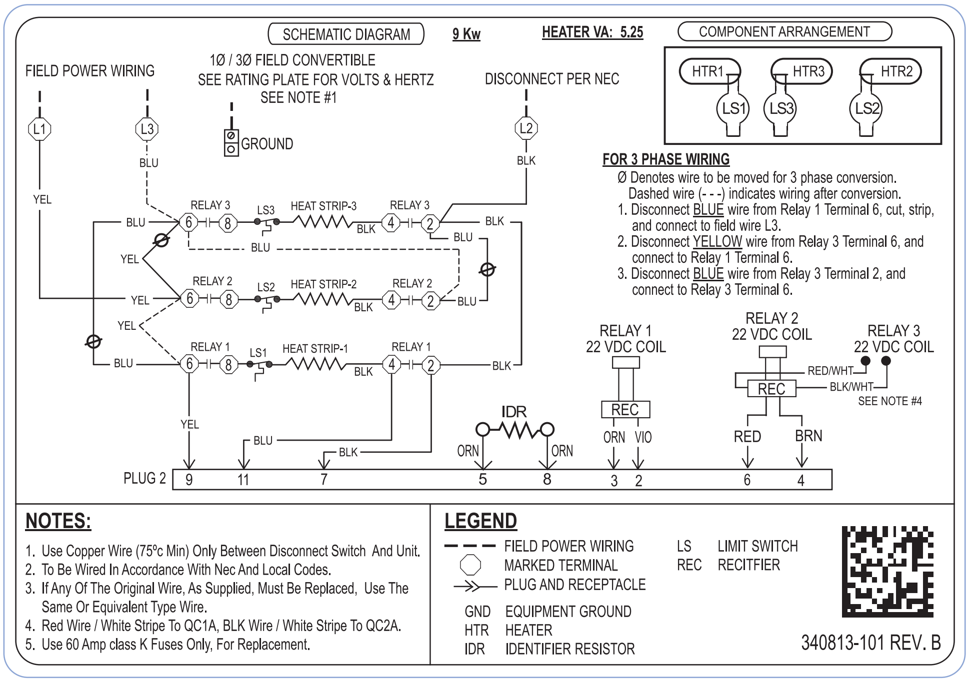
Fig. 1 - 340816-101

A150105



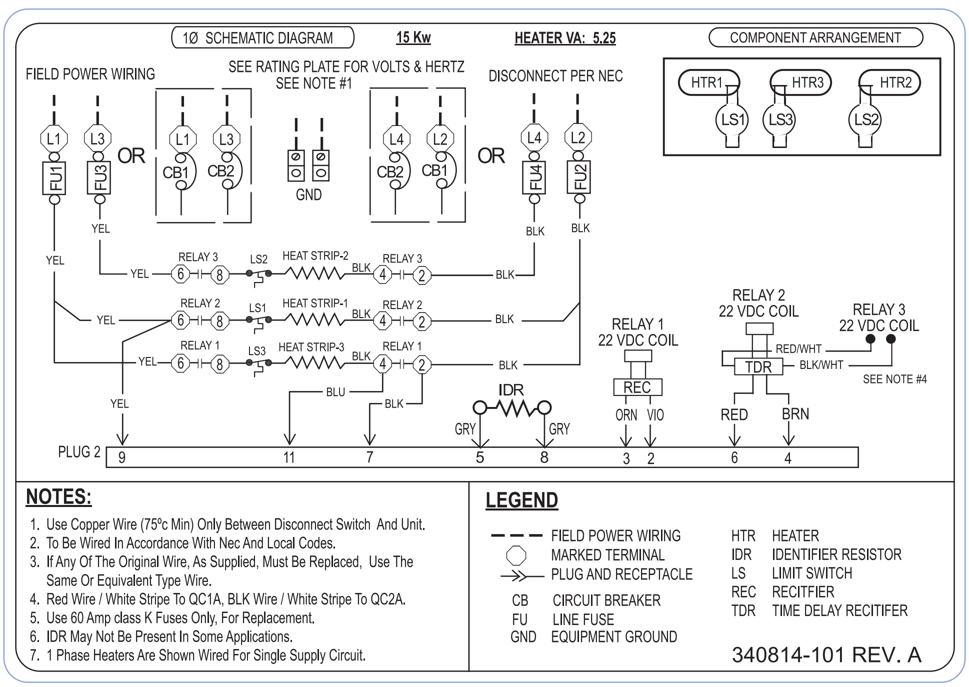
A150106

Fig. 2 - 340817-101



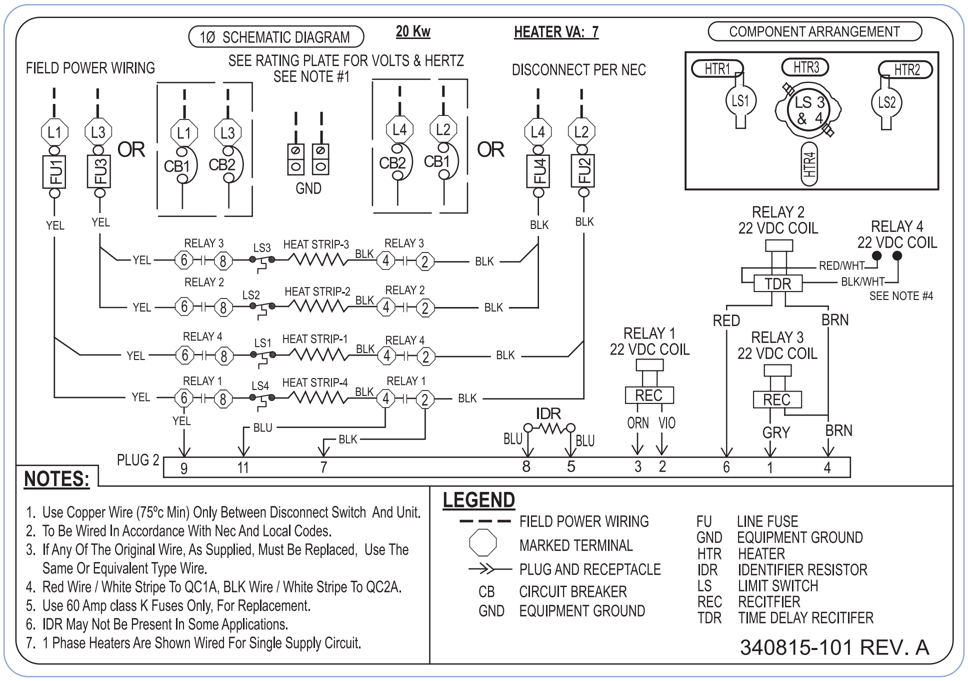
A150102

Fig. 3 - 340813-101



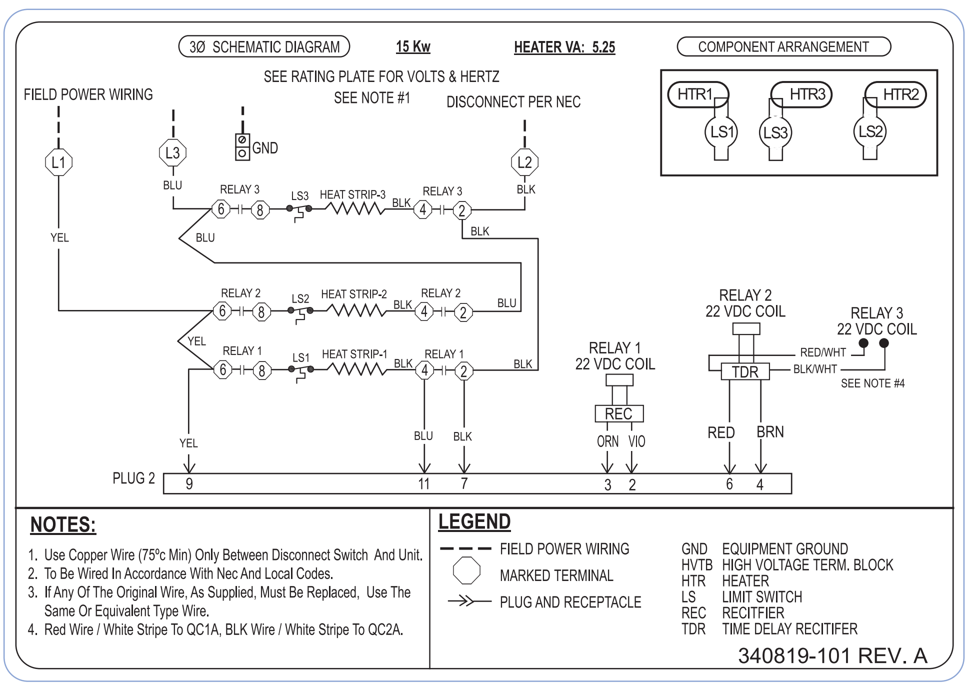
A150103

Fig. 4 - 340814-101



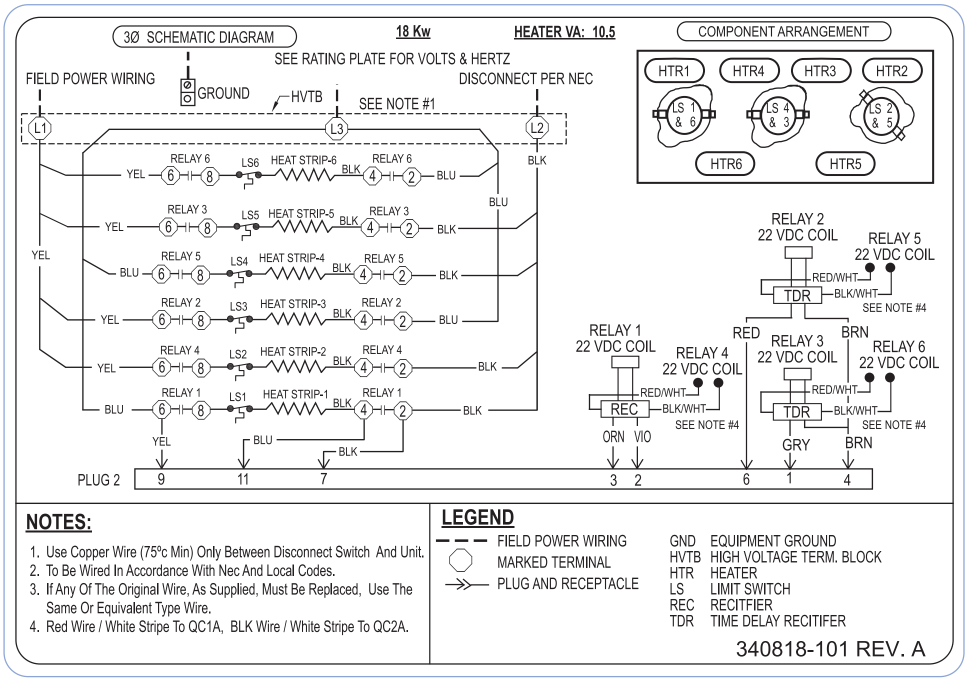
A150104

Fig. 5 - 340815-101



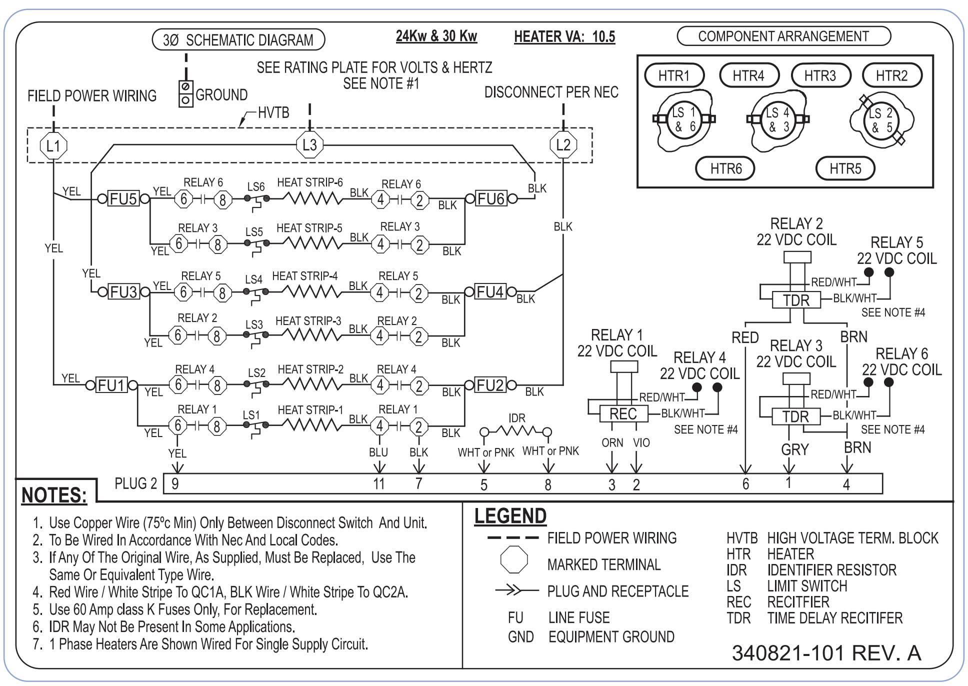
A150108

Fig. 6 - 340819-101



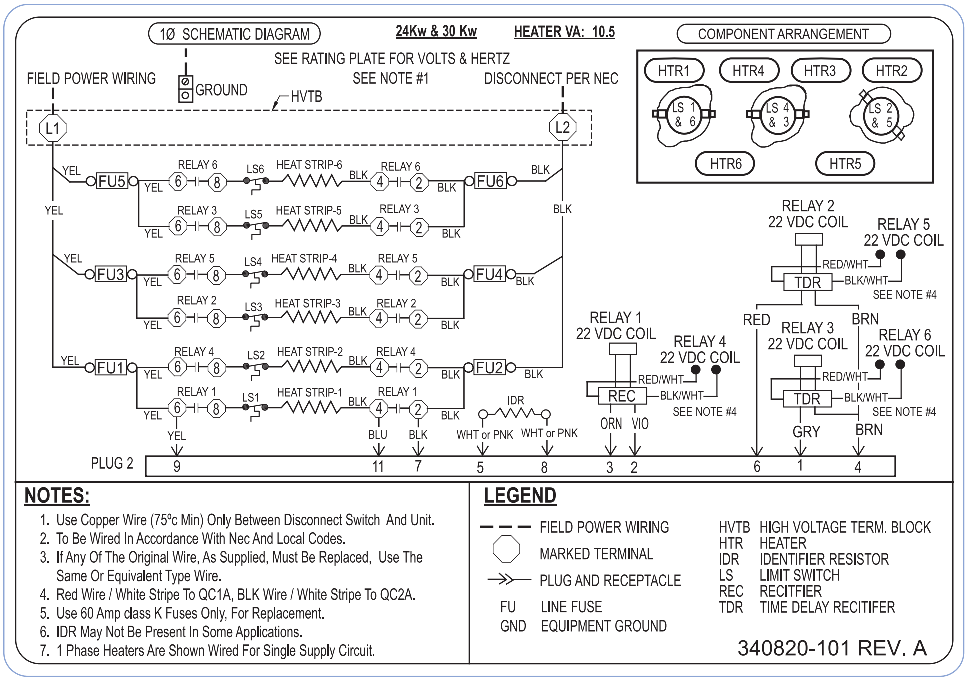
A150107

Fig. 7 - 340818-101



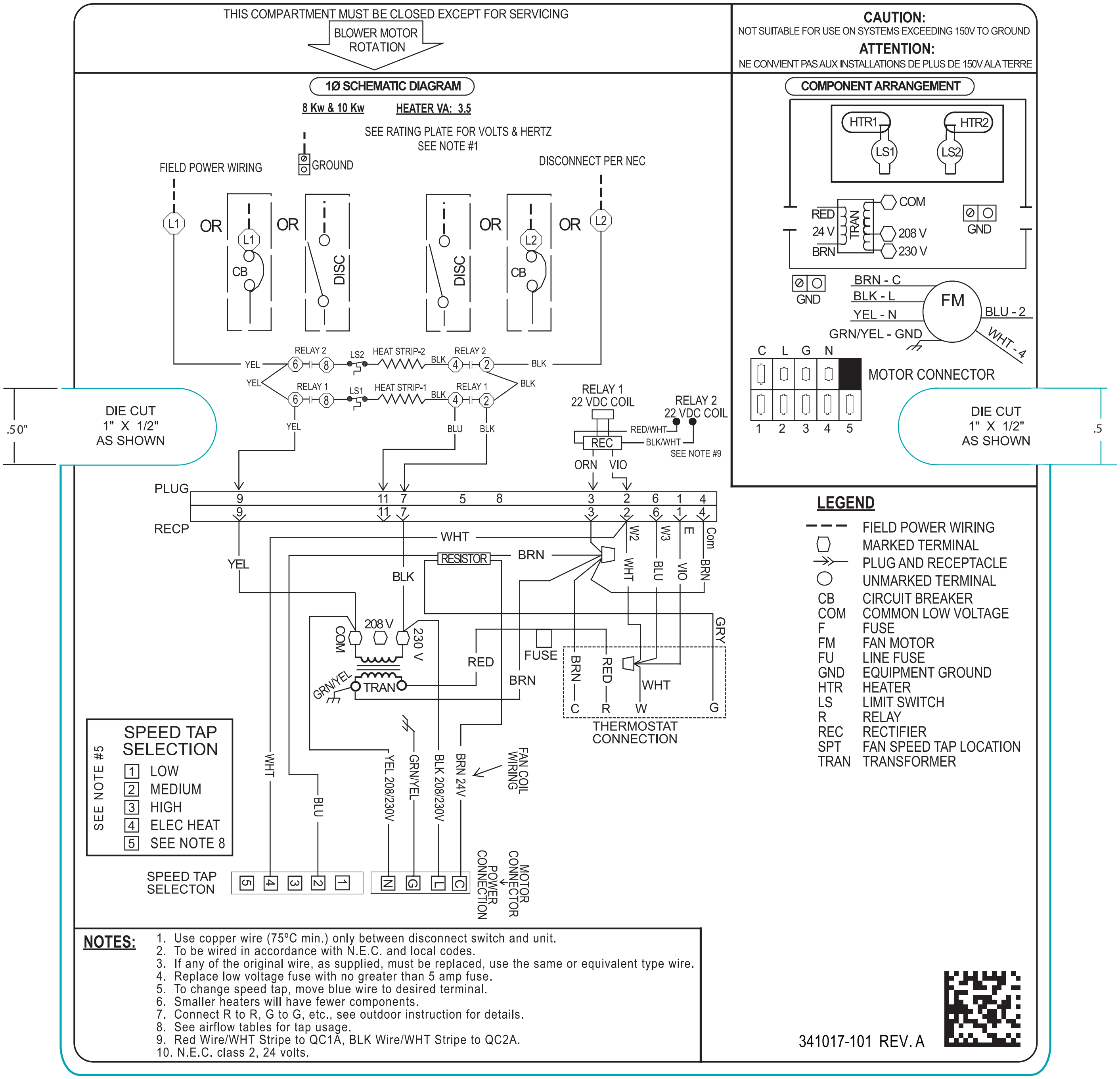
A150109

Fig. 8 - 340821-101



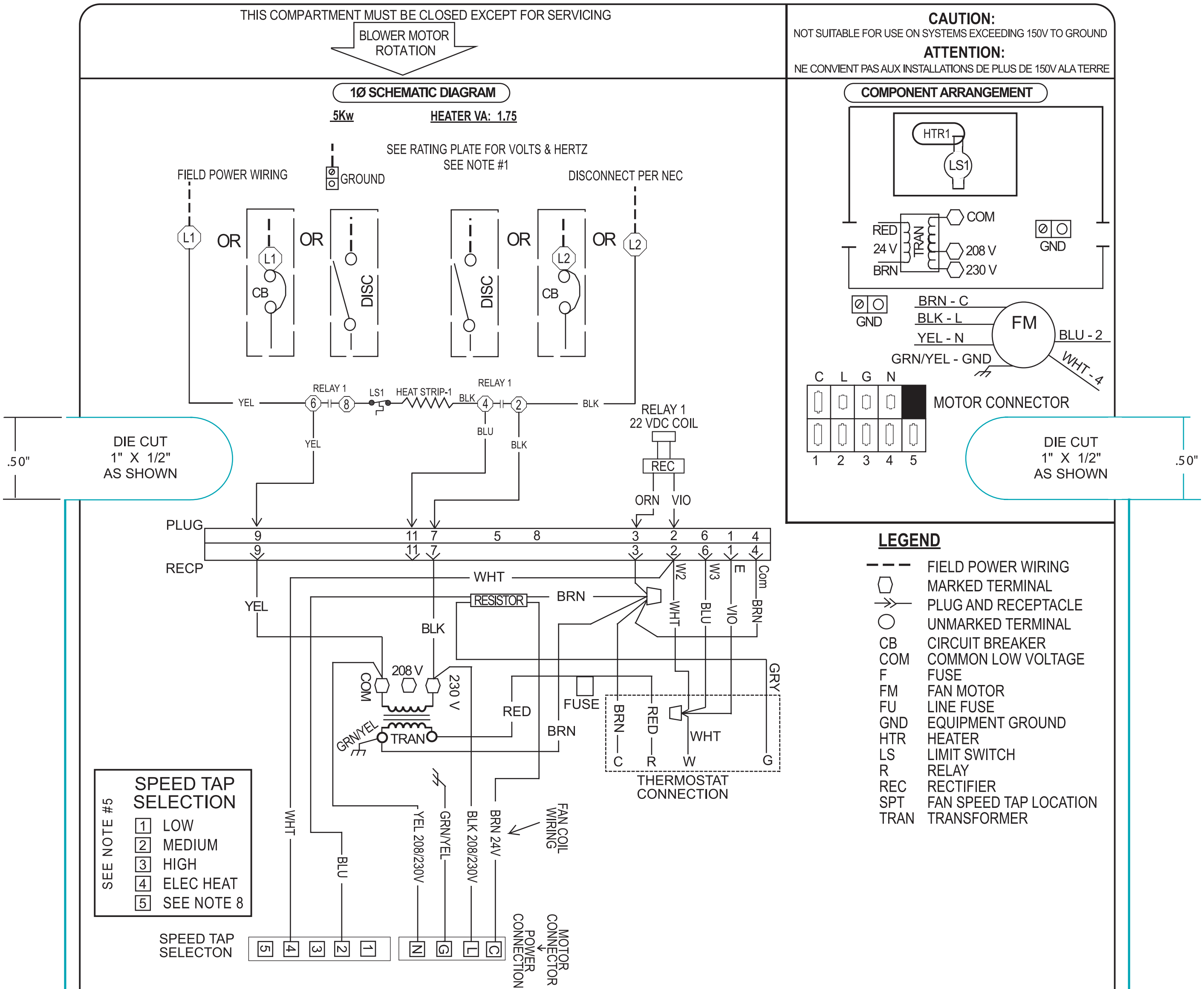
A150110

Fig. 9 - 340820-101



A150111

Fig. 10 - 341017-101



NOTES:

1. Use copper wire (75°C min.) only between disconnect switch and unit.
2. To be wired in accordance with N.E.C. and local codes.
3. If any of the original wire, as supplied, must be replaced, use the same or equivalent type wire.
4. Replace low voltage fuse with no greater than 5 amp fuse.
5. To change speed tap, move blue wire to desired terminal.
6. Smaller heaters will have fewer components.
7. Connect R to R, G to G, etc., see outdoor instruction for details.
8. See airflow tables for tap usage.
9. N.E.C. class 2, 24 volts.

341018-101 REV. A

A150112

Fig. 11 - 341018-101

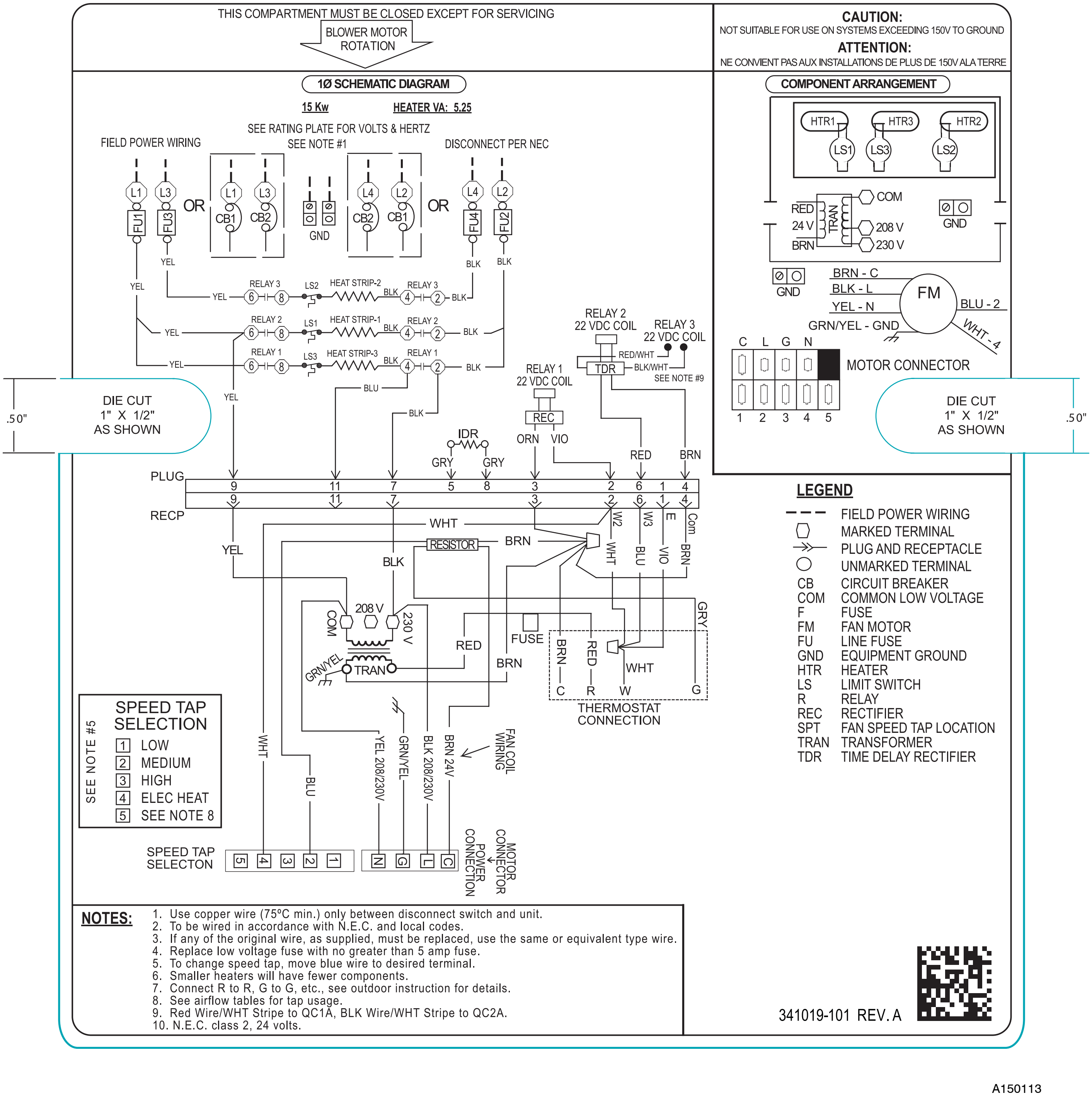
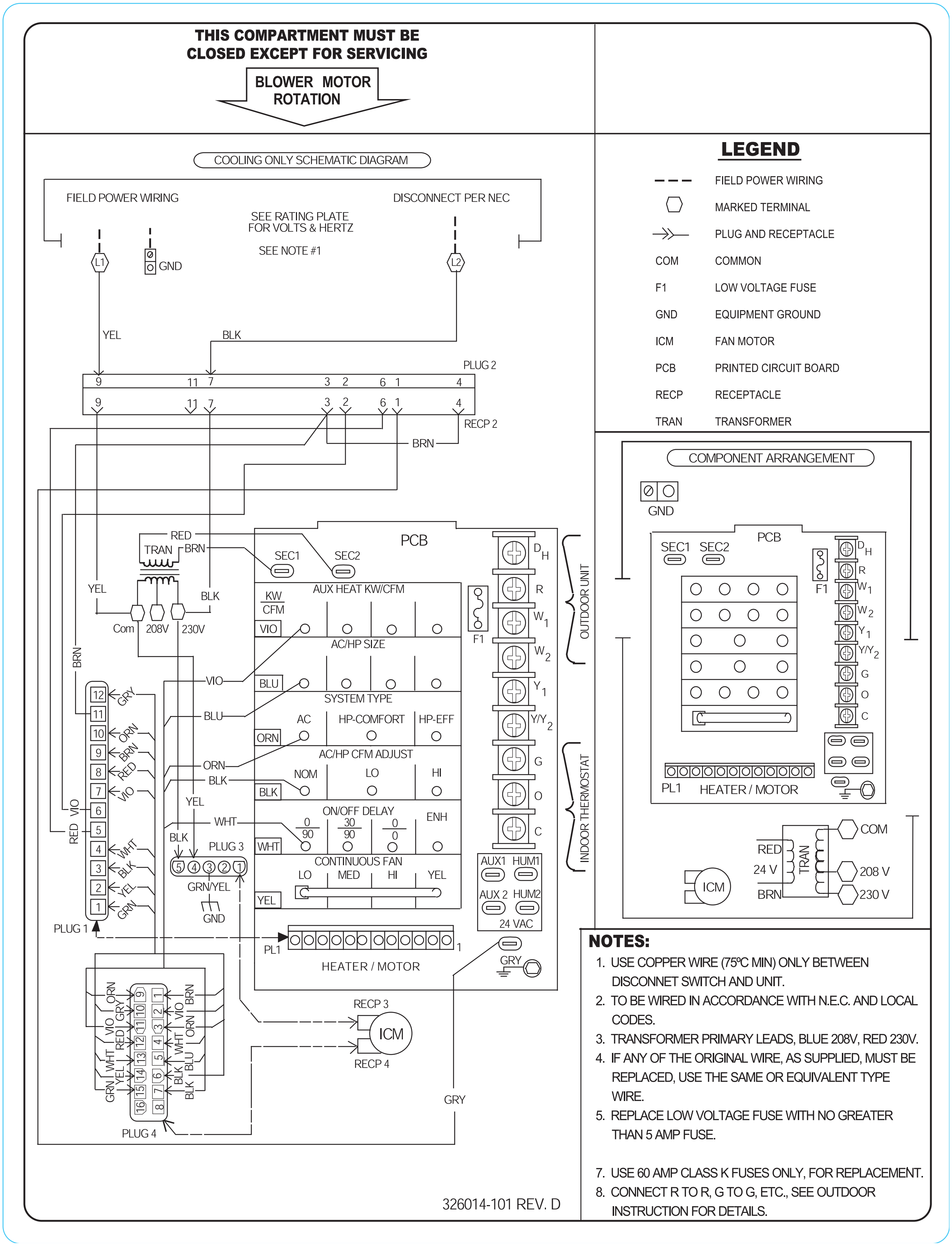
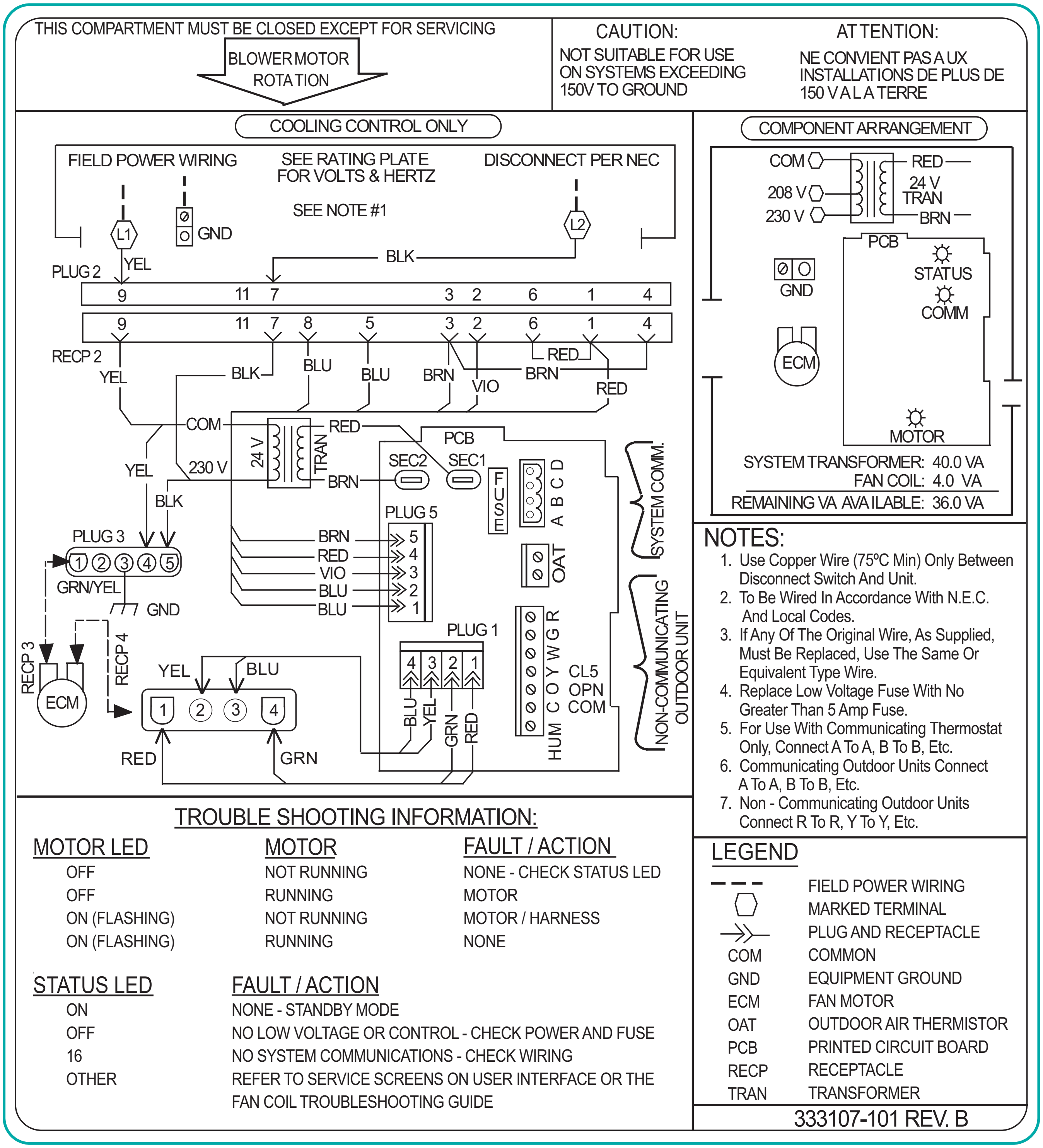


Fig. 12 - 341019-101



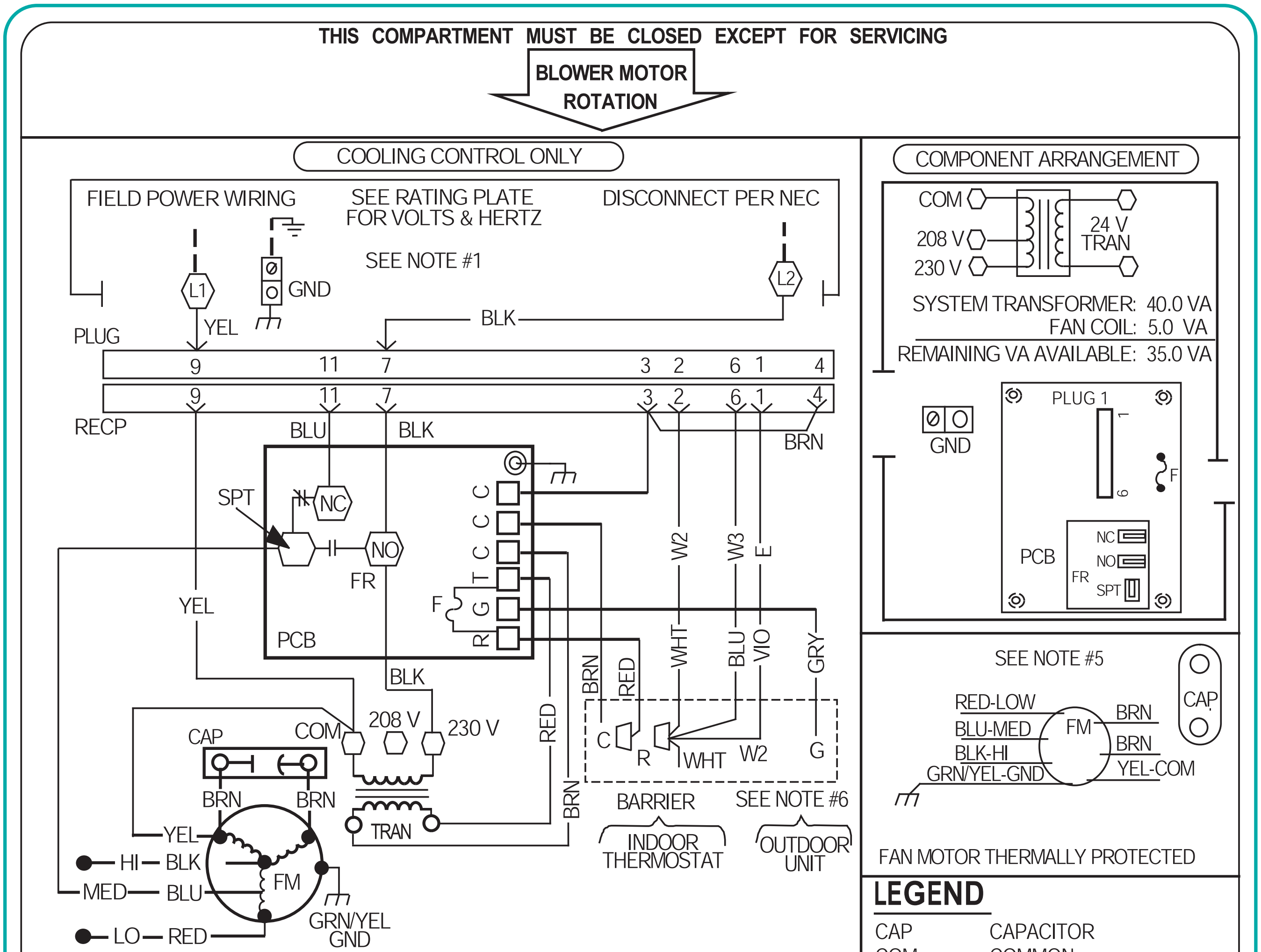
A07029

Fig. 13 - 326014-101



A12363

Fig. 14 - 333107-101



Minimum Motor Speed Tap Selection For Electric Heater

MODEL SIZE	HEATER SIZE KW 3,5,8	9	10	15	18	20	24, 30
18	MED *	-----	HI	-----	-----	-----	-----
24	MED‡	-----	MED‡	MED‡	-----	-----	-----
30	-----	-----	LO	LO	-----	MED *	----
36, 42, 48, 60	-----	LO	LO	LO	LO	LO	LO
70	-----	MED	MED	MED	MED	MED	MED

* - MED speed on 3 speed motors and HI speed on 2 speed motors.
‡ - MED speed on 3 speed motors and LO speed on 2 speed motors.

NOTES

1. Use Copper Wire (75ºC Min) Only Between Disconnect Switch And Unit.
2. To Be Wired In Accordance With NEC And Local Codes.
3. If Any Of The Original Wire, As Supplied, Must Be Replaced, Use The Same Or Equivalent Type Wire.
4. Replace Low Voltage Fuse With No Greater Than 5 Amp Fuse.
5. (3) Speed Motor Shown Optional (2) Speed Motor Uses HI (BLK) And LOW (BLUE or RED).
6. Connect R To R, G To G, Etc. See Outdoor Instruction For Details.

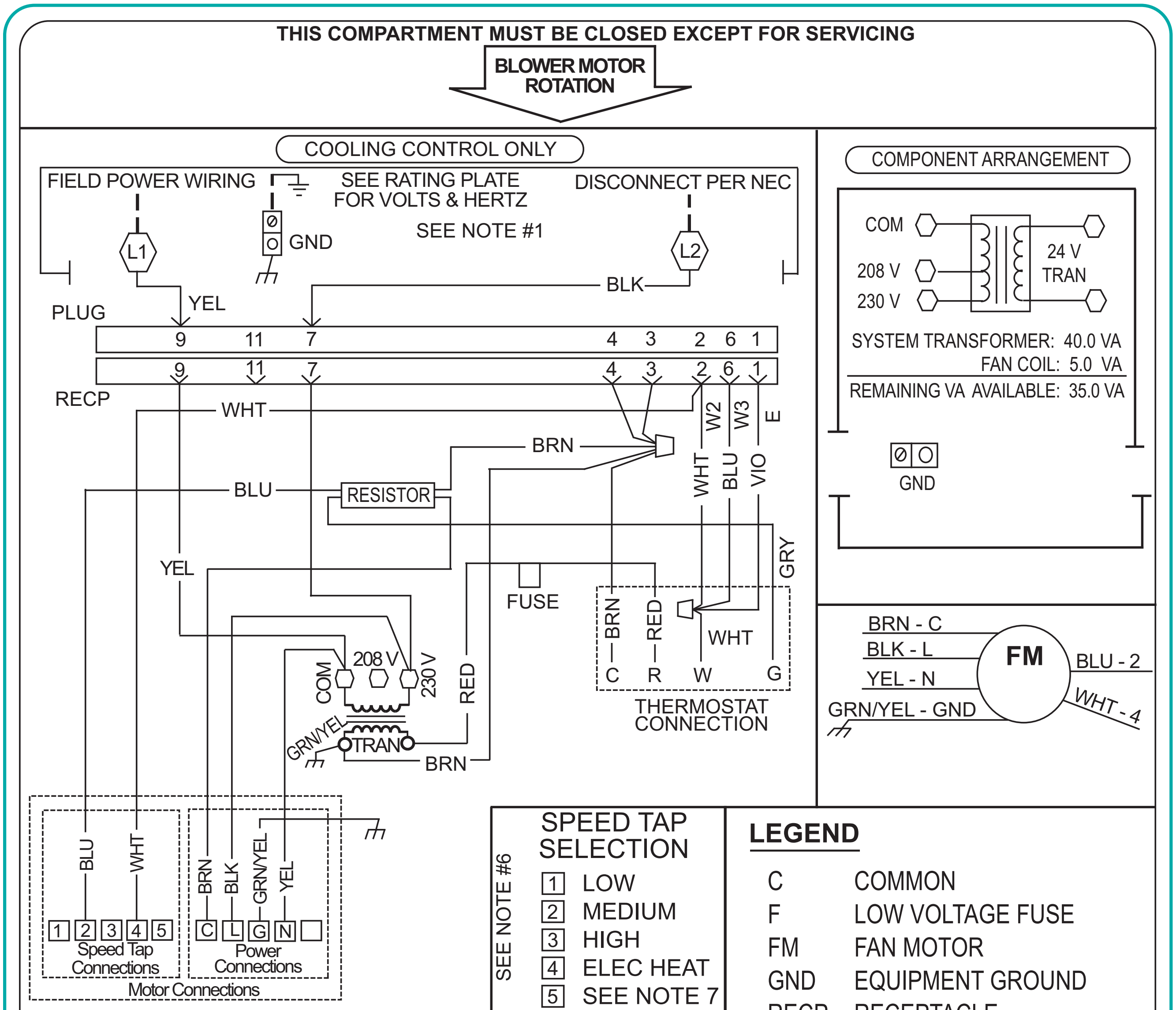
LEGEND

CAP	CAPACITOR
COM	COMMON
F	LOW VOLTAGE FUSE
FR	PCB FAN RELAY
FM	FAN MOTOR
GND	EQUIPMENT GROUND
PCB	PRINTED CIRCUIT BOARD
RECP	RECEPTACLE
SPT	FAN SPEED TAP LOCATION
TRAN	TRANSFORMER
○	UNMARKED TERMINAL
- - -	FIELD POWER WIRING
⬡	MARKED TERMINAL
→≻	PLUG AND RECEPTACLE

328964-101 REV. A

A07027

Fig. 15 - 328964-101



SEE NOTE #6

SPEED TAP SELECTION

1 LOW
2 MEDIUM
3 HIGH
4 ELEC HEAT
5 SEE NOTE 7

NOTES

1. Use Copper Wire (75ºC Min) Only Between Disconnect Switch And Unit.
2. To Be Wired In Accordance With NEC And Local Codes.
3. If Any Of The Original Wire, As Supplied, Must Be Replaced. Use The Same Or Equivalent Type Wire.
4. Replace Low Voltage Fuse With No Greater Than 5 Amp Fuse.
5. Connect R To R, G To G, Etc. See Outdoor Instruction For Details.
6. To change speed tap, move blue wire to desired terminal.
7. See airflow tables for tap usage.

LEGEND

C	COMMON
F	LOW VOLTAGE FUSE
FM	FAN MOTOR
GND	EQUIPMENT GROUND
RECP	RECEPTACLE
SPT	FAN SPEED TAP LOCATION
TRAN	TRANSFORMER
O	UNMARKED TERMINAL
- - -	FIELD POWER WIRING
⬡	MARKED TERMINAL
→≻	PLUG AND RECEPTACLE

336228-101 REV. A

A12364

Fig. 16 - 336228-101

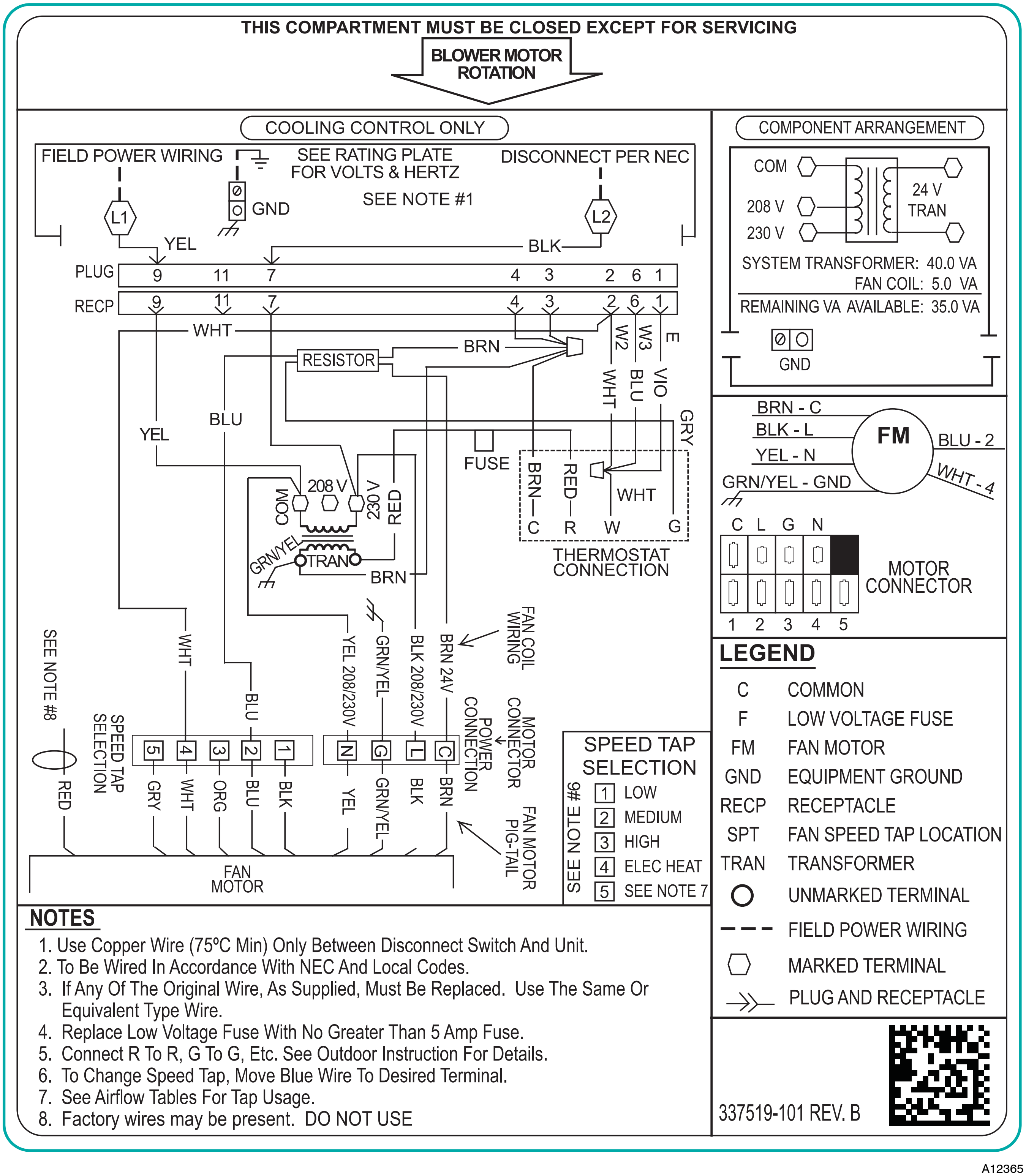
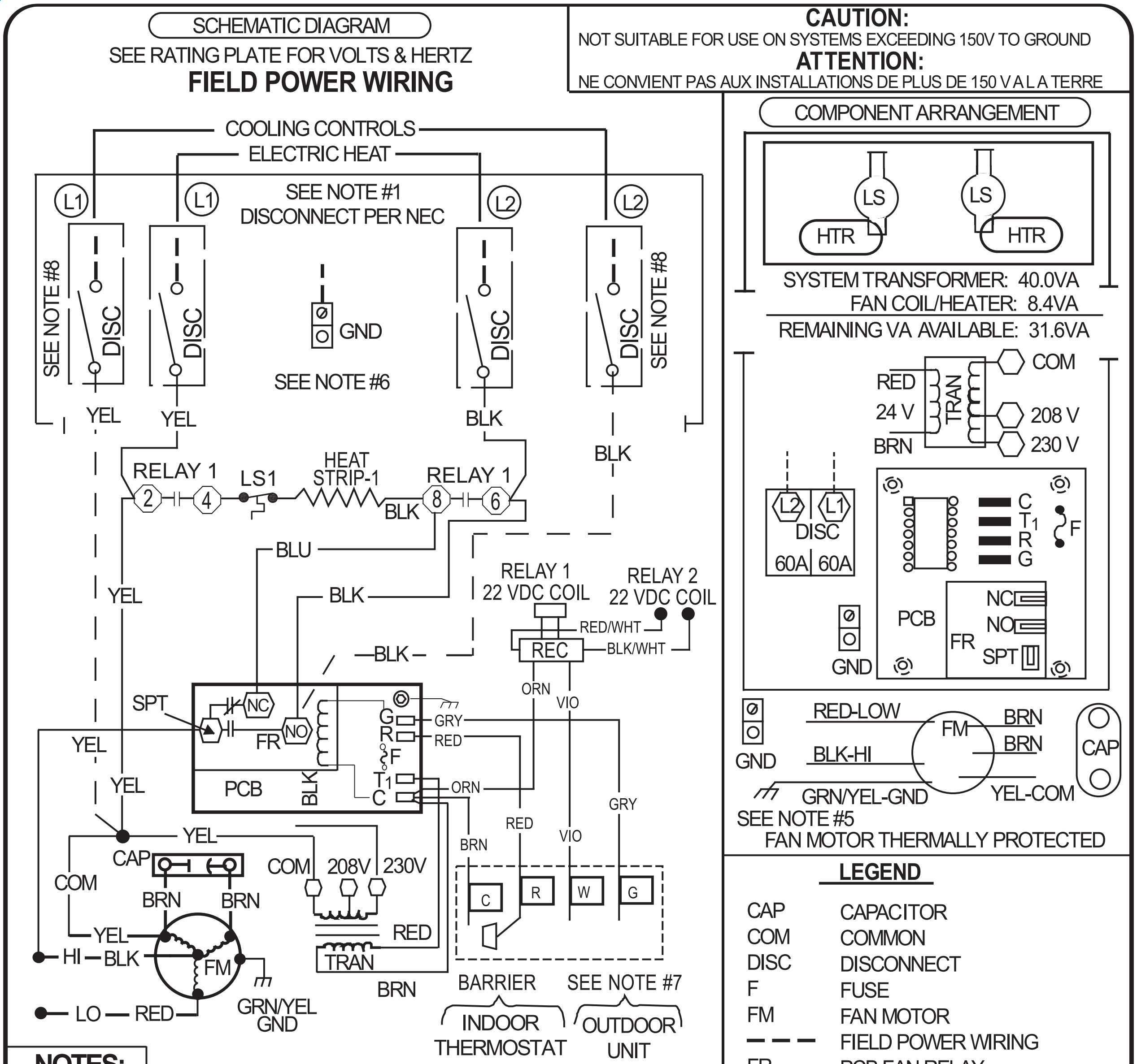


Fig. 17 - 337519-101



NOTES:

1. Use copper wire (75º C min) only between disconnect switch and unit.
2. To be wired in accordance with N.E.C. and local codes.
3. If any of the original wire, as supplied, must be replaced, use the same or equivalent type wire.
4. Replace low voltage fuse with no greater than 5 AMP fuse.
5. (2) Speed Motor uses HI (BLK) and LOW (RED).
6. Smaller heaters will have fewer components.
7. Connect R to R, G to G, etc., see outdoor instructions for details.
8. Cooling controls wiring not used with electric heaters.
9. Motor speed selection for 018 and 024 sizes with 11 kW heater and heat pump is High. All other heater combinations is Low.
10. N.E.C. Class 2, 24 Volts.

LEGEND

CAP	CAPACITOR
COM	COMMON
DISC	DISCONNECT
F	FUSE
FM	FAN MOTOR
– – –	FIELD POWER WIRING
FR	PCB FAN RELAY
GND	EQUIPMENT GROUND
HTR	HEATER
LS	LIMIT SWITCH
⬡	MARKED TERMINAL
→>—	PLUG AND RECEPTACLE
PCB	PRINTED CIRCUIT BOARD
REC	RECTIFIER
SPT	FAN SPEED TAP LOCATION
TRAN	TRANSFORMER
○	UNMARKED TERMINAL

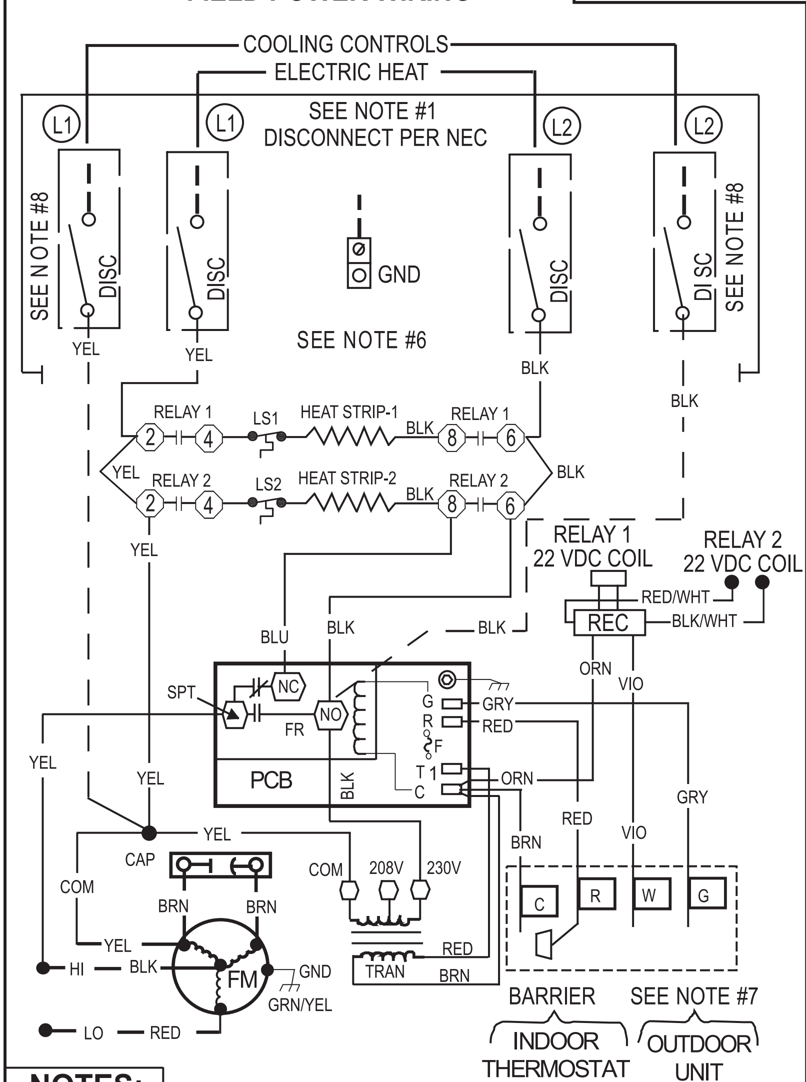
This compartment must be closed except when servicing.

ROTATION

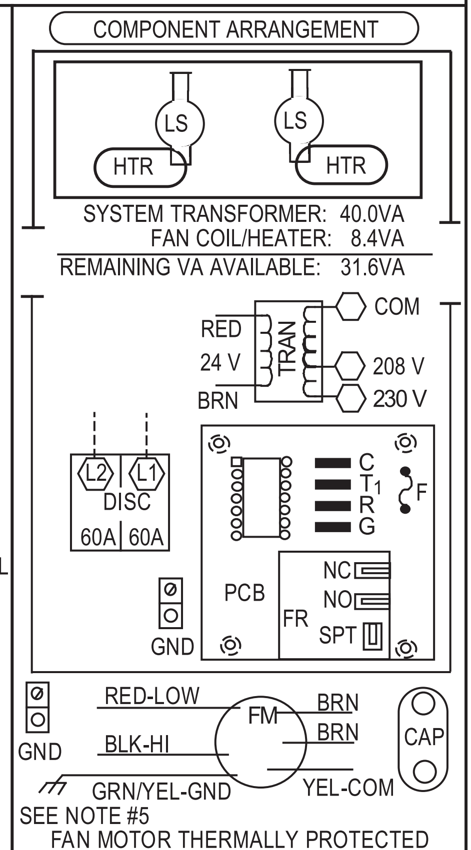
341080-101 REV. A

A150114

Fig. 18 - 341080-101



CAUTION:
NOT SUITABLE FOR USE ON SYSTEMS EXCEEDING 150V TO GROUND
ATTENTION:
NE CONVIENT PAS AUX INSTALLATIONS DE PLUS DE 150V ALA TERRE



NOTES:

1. Use copper wire (75º C min) only between disconnect switch and unit.
2. To be wired in accordance with N.E.C. and local codes.
3. If any of the original wire, as supplied, must be replaced, use the same or equivalent type wire.
4. Replace low voltage fuse with no greater than 5 AMP fuse.
5. (2) Speed Motor uses HI (BLK) and LOW (RED).
6. Smaller heaters will have fewer components.
7. Connect R to R, G to G, etc., see outdoor instructions for details.
8. Cooling controls wiring not used with electric heaters.
9. Motor speed selection for 018 and 024 sizes with 11 kW heater and heat pump is High. All other heater combinations is Low.
10. N.E.C. Class 2, 24 Volts.

LEGEND

CAP	CAPACITOR
COM	COMMON
DISC	DISCONNECT
F	FUSE
FM	FAN MOTOR
— — —	FIELD POWER WIRING
FR	PCB FAN RELAY
GND	EQUIPMENT GROUND
HTR	HEATER
LS	LIMIT SWITCH
⬡	MARKED TERMINAL
→≫—	PLUG AND RECEPTACLE
PCB	PRINTED CIRCUIT BOARD
REC	RECTIFIER
SPT	FAN SPEED TAP LOCATION
TRAN	TRANSFORMER
○	UNMARKED TERMINAL

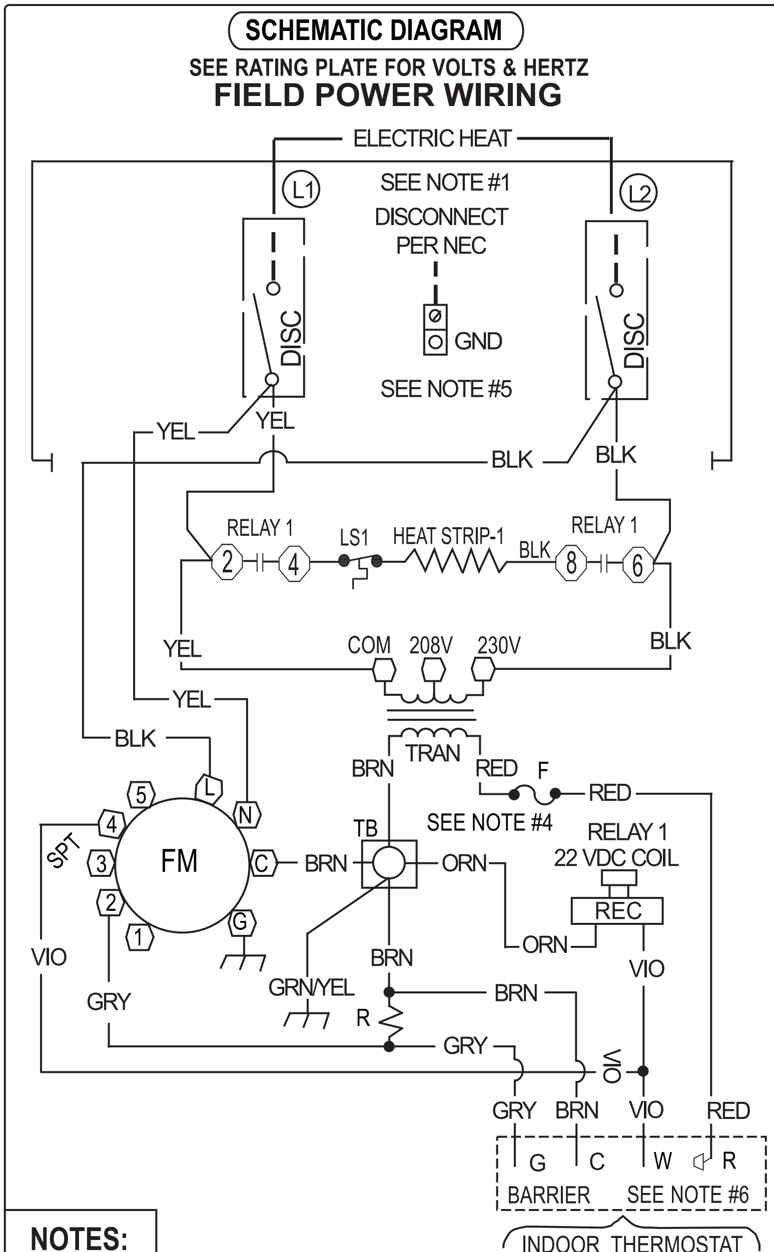
This compartment must be closed except when servicing.

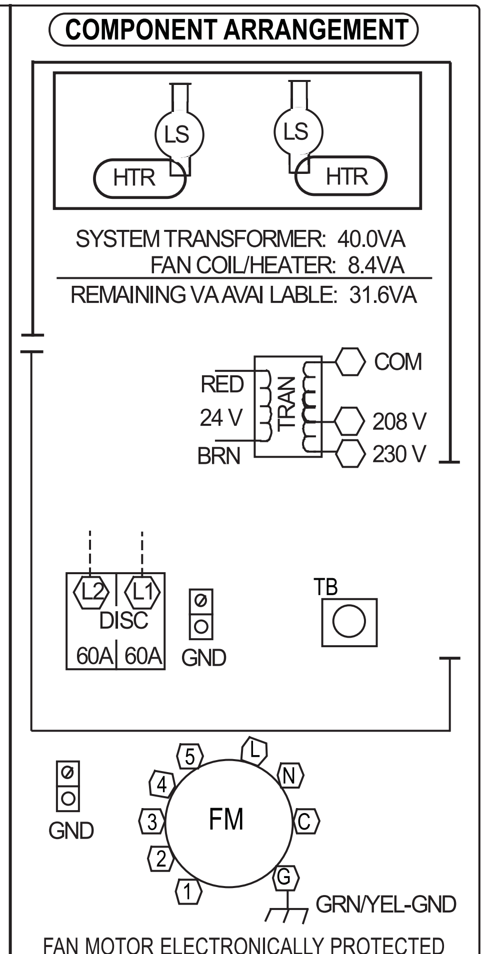
← ROTATION

341081-101 REV. A

A150115

Fig. 19 - 341081-101





LEGEND

— — —	FIELD POWER WIRING
⬡	MARKED TERMINAL
→≻	PLUG AND RECEPTACLE
○	UNMARKED TERMINAL
	WIRE NUT
COM	COMMON
DISC	DISCONNECT
F	FUSE
FM	FAN MOTOR
GND	EQUIPMENT GROUND
HIR	HEATER
LS	LIMIT SWITCH
R	RESISTOR
REC	RECTIFIER
SPT	FAN SPEED TAP LOCATION
TB	TERMINAL BLOCK
TRAN	TRANSFORMER

NOTES:

1. Use copper wire (75º C min) only between disconnect switch and unit.
2. To be wired in accordance with N.E.C. and local codes.
3. If any of the original wire, as supplied, must be replaced, use the same or equivalent type wire.
4. Replace low voltage fuse with no greater than 3 AMP fuse.
5. Smaller heaters will have fewer components.
6. Connect R to R, G to G, etc., see outdoor instructions for details.
7. Cooling speed selection can be tap 1, 2, 3, or 5.
8. Heating speed selection must be tap 4 only.
9. N.E.C. class 2, 24 volts.

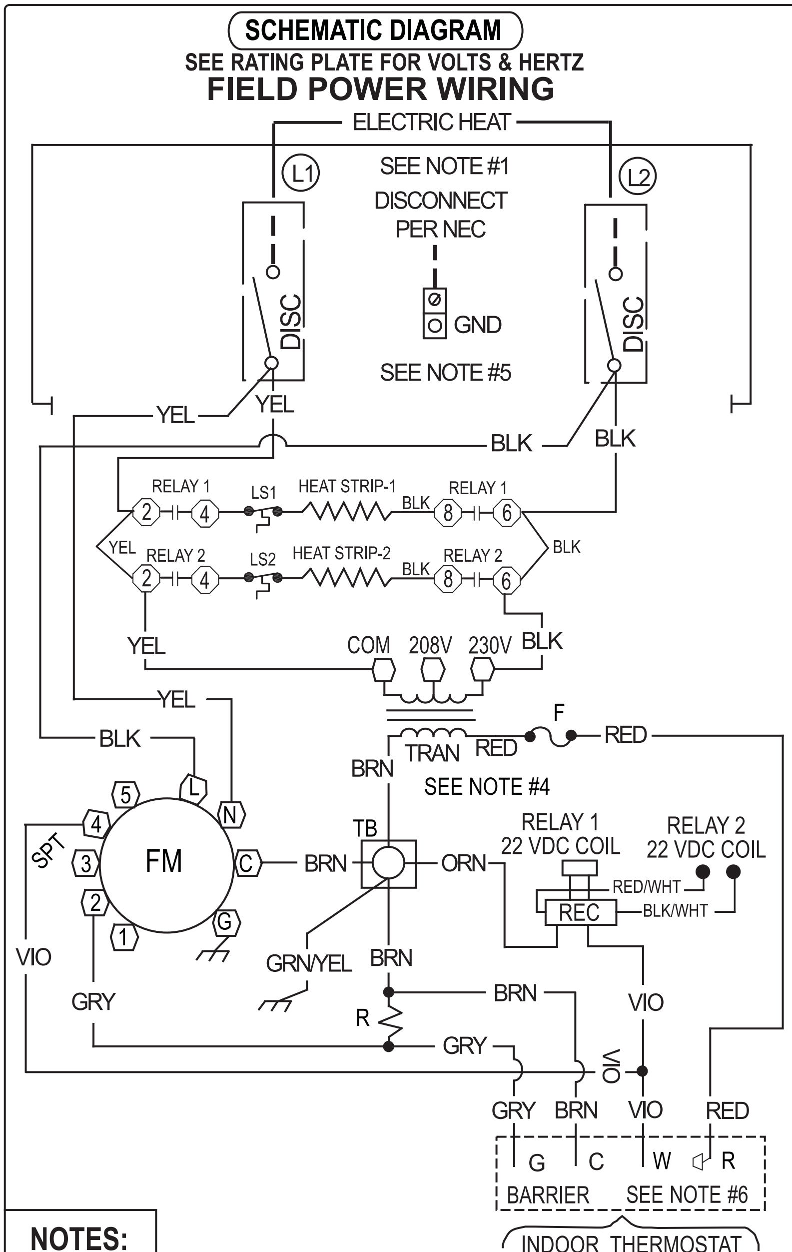
This compartment must be closed except when servicing.

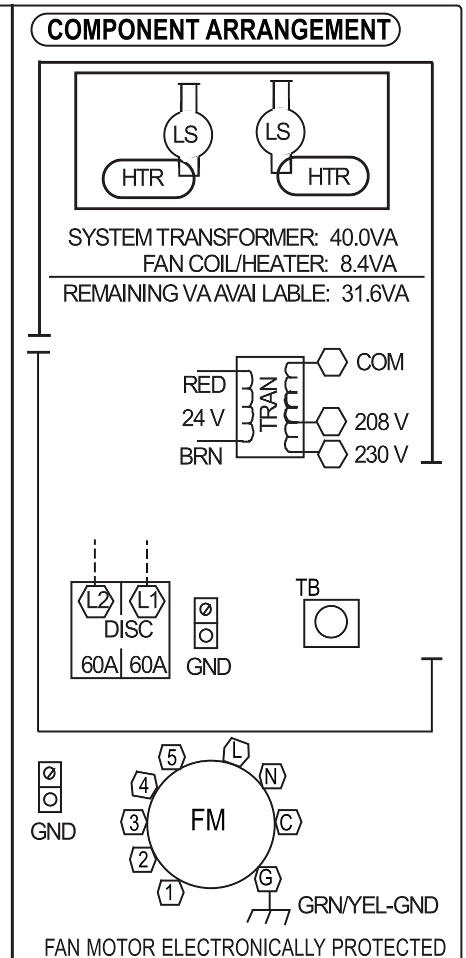
ROTATION

341082-101 REV. A

A150116

Fig. 20 - 341082-101





LEGEND

— — —	FIELD POWER WIRING
⬡	MARKED TERMINAL
→≻—	PLUG AND RECEPTACLE
○	UNMARKED TERMINAL
	WIRE NUT
COM	COMMON
DISC	DISCONNECT
F	FUSE
FM	FAN MOTOR
GND	EQUIPMENT GROUND
HIR	HEATER
LS	LIMIT SWITCH
R	RESISTOR
REC	RECTIFIER
SPT	FAN SPEED TAP LOCATION
TB	TERMINAL BLOCK
TRAN	TRANSFORMER

NOTES:

1. Use copper wire (75º C min) only between disconnect switch and unit.
2. To be wired in accordance with N.E.C. and local codes.
3. If any of the original wire, as supplied, must be replaced, use the same or equivalent type wire.
4. Replace low voltage fuse with no greater than 3 AMP fuse.
5. Smaller heaters will have fewer components.
6. Connect R to R, G to G, etc., see outdoor instructions for details.
7. Cooling speed selection can be tap 1, 2, 3, or 5.
8. Heating speed selection must be tap 4 only.
9. N.E.C. class 2, 24 volts.

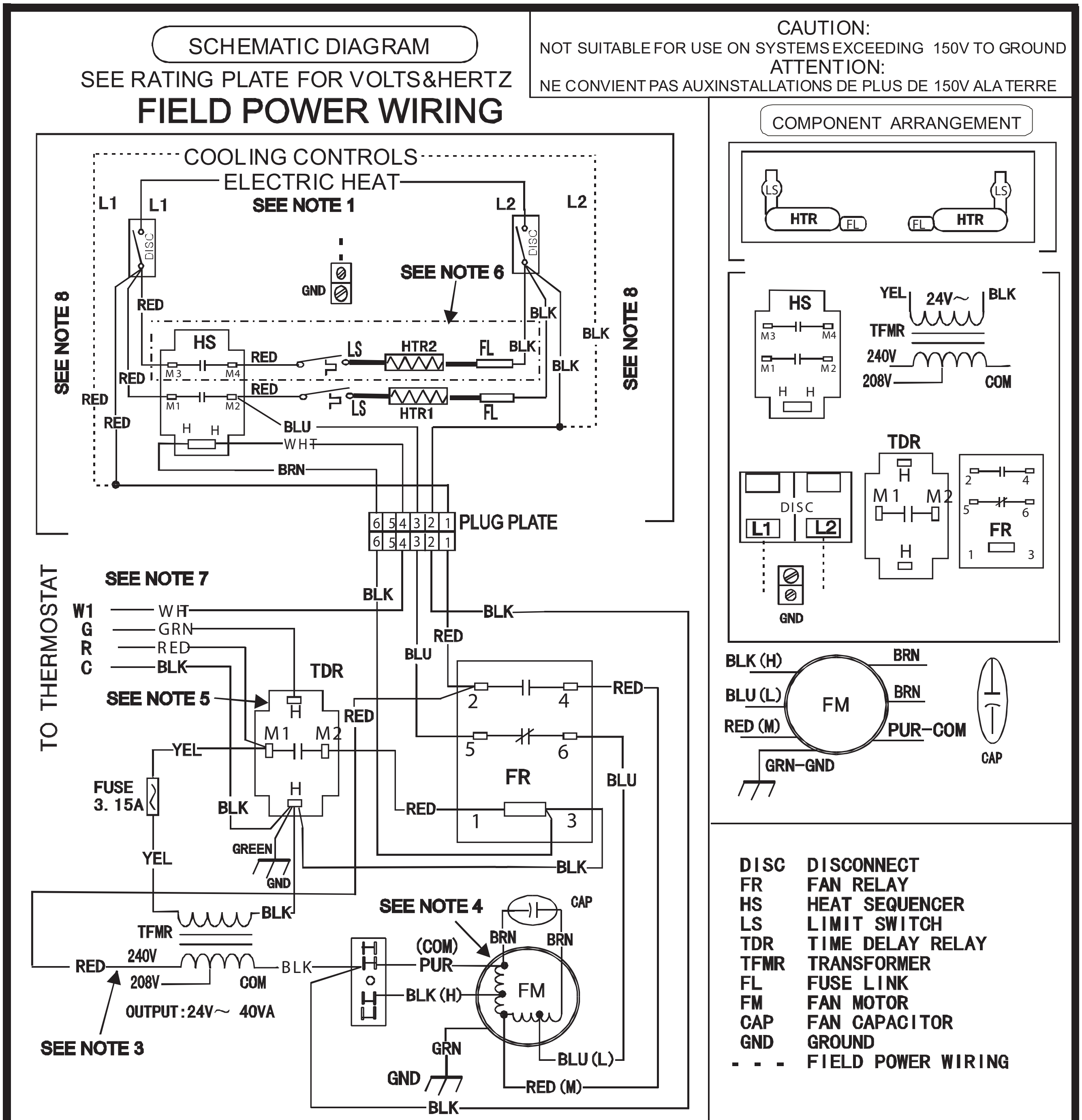
This compartment must be closed except when servicing.

ROTATION

341083-101 REV. A

A150117

Fig. 21 - 341083-101



NOTES:

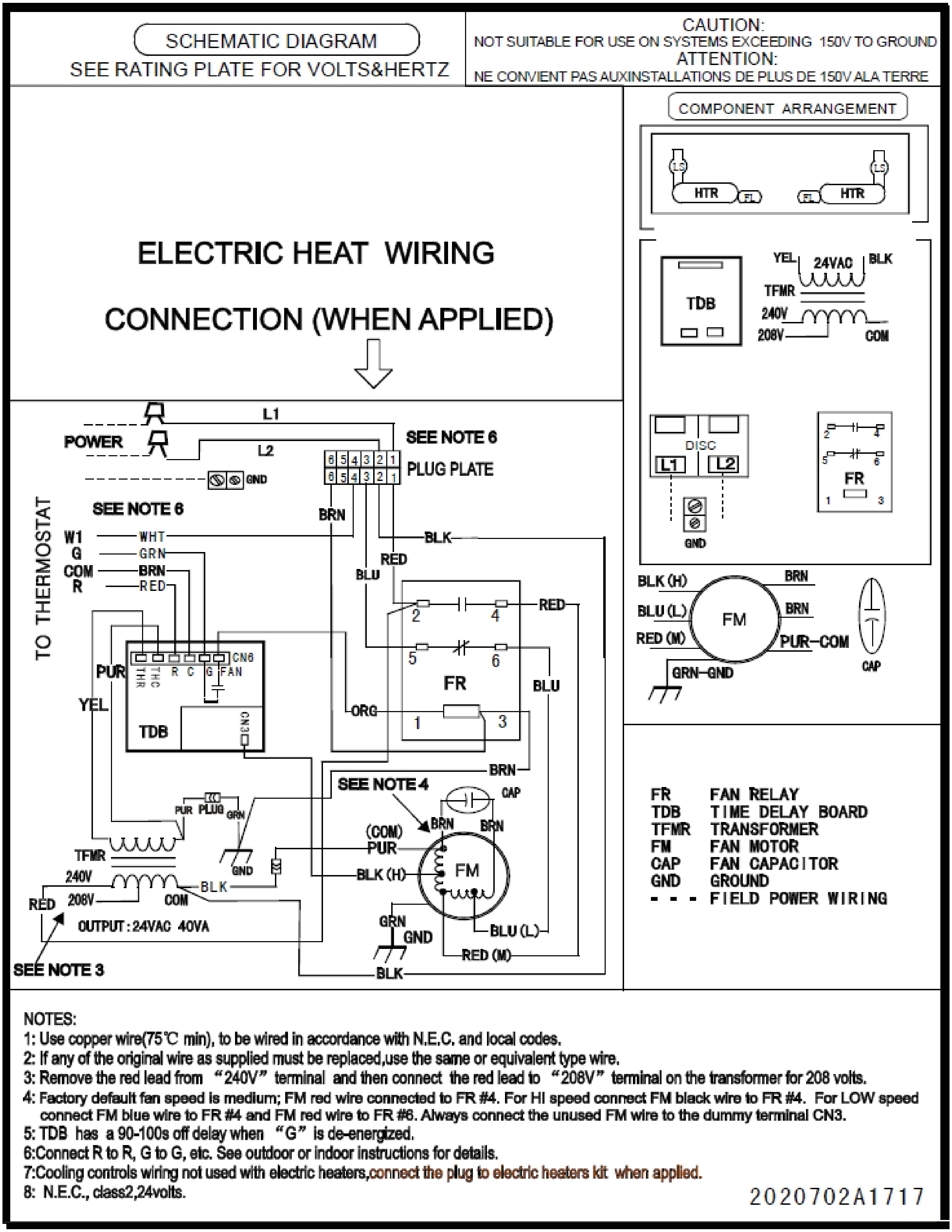
1: Use copper wire(75℃ min) only between disconnect switch and unit, To be wired in accordance with N.E.C. and local codes.
2: If any of the original wire as supplied must be replaced,use the same or equivalent type wire.
3: Remove the red lead from "240V" terminal and then connect the red lead to "208V" terminal on the transformer for 208 volts.
4: Factory default fan speed is medium; FM red wire connected to FR #4. For HI speed connect FM black wire to FR #4. For LOW speed connect FM blue wire to FR #4 and FM red wire to FR #6. Always connect the unused FM wire to the dummy terminal block.
5: TDR has a 1-30s on delay when "G" is energized and a 45-75s off delay when "G" is de-energized.
6: The 5kW heater kit has HTR1 only. Fan coils equipped with electric heat connect power supply to circuit breaker.
7:Connect R to R, G to G, etc. See outdoor or indoor instructions for details.
8:Cooling controls wiring not used with electric heaters.

202070290385

A13131

Fig. 22 – FFMANP(018,024,030,036 and EHK2 Electric Heaters with Sequencers

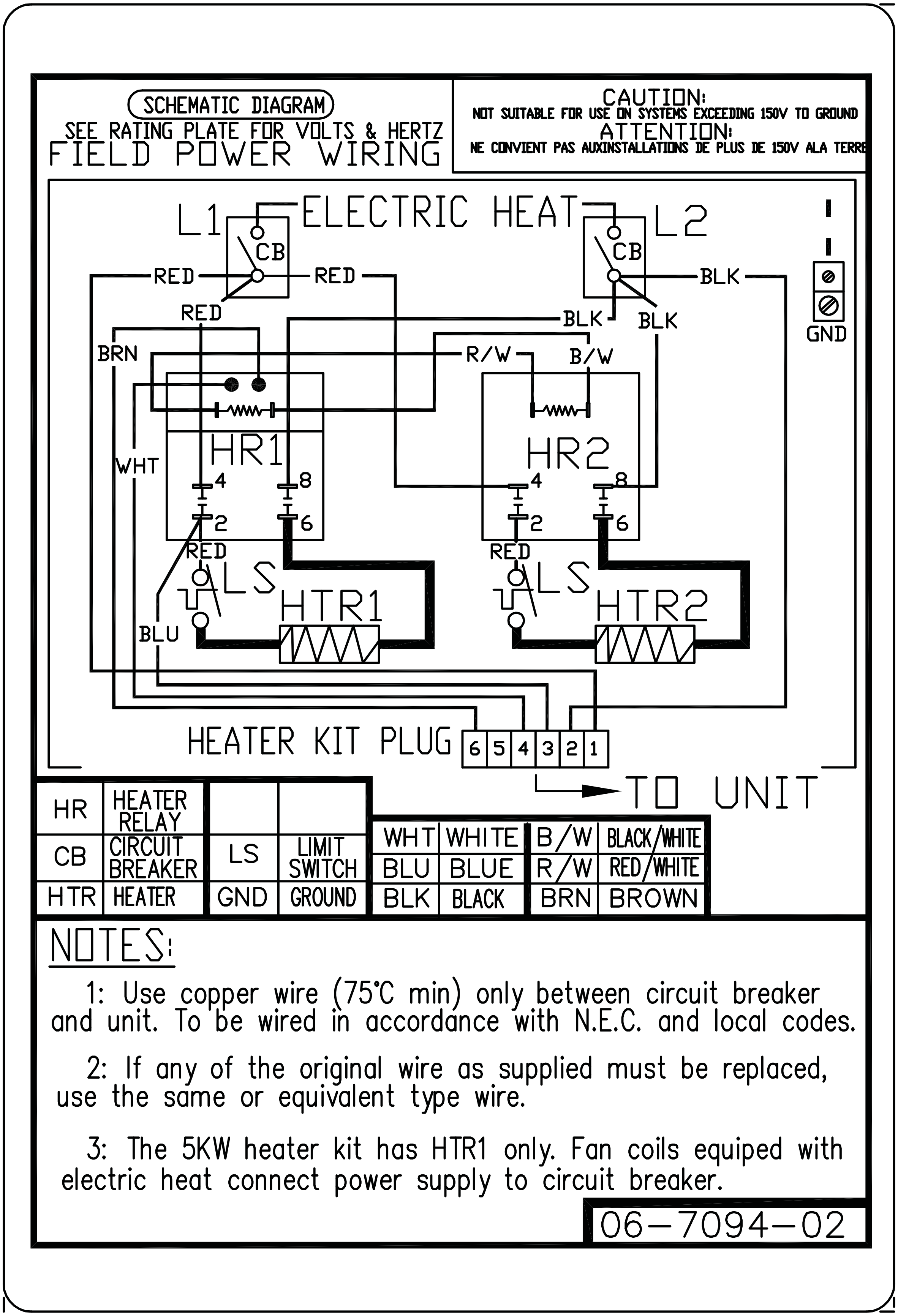
NOTE: Representative for FFMANP(018,024,030,036) prior to serial number date code 1715V



A150153

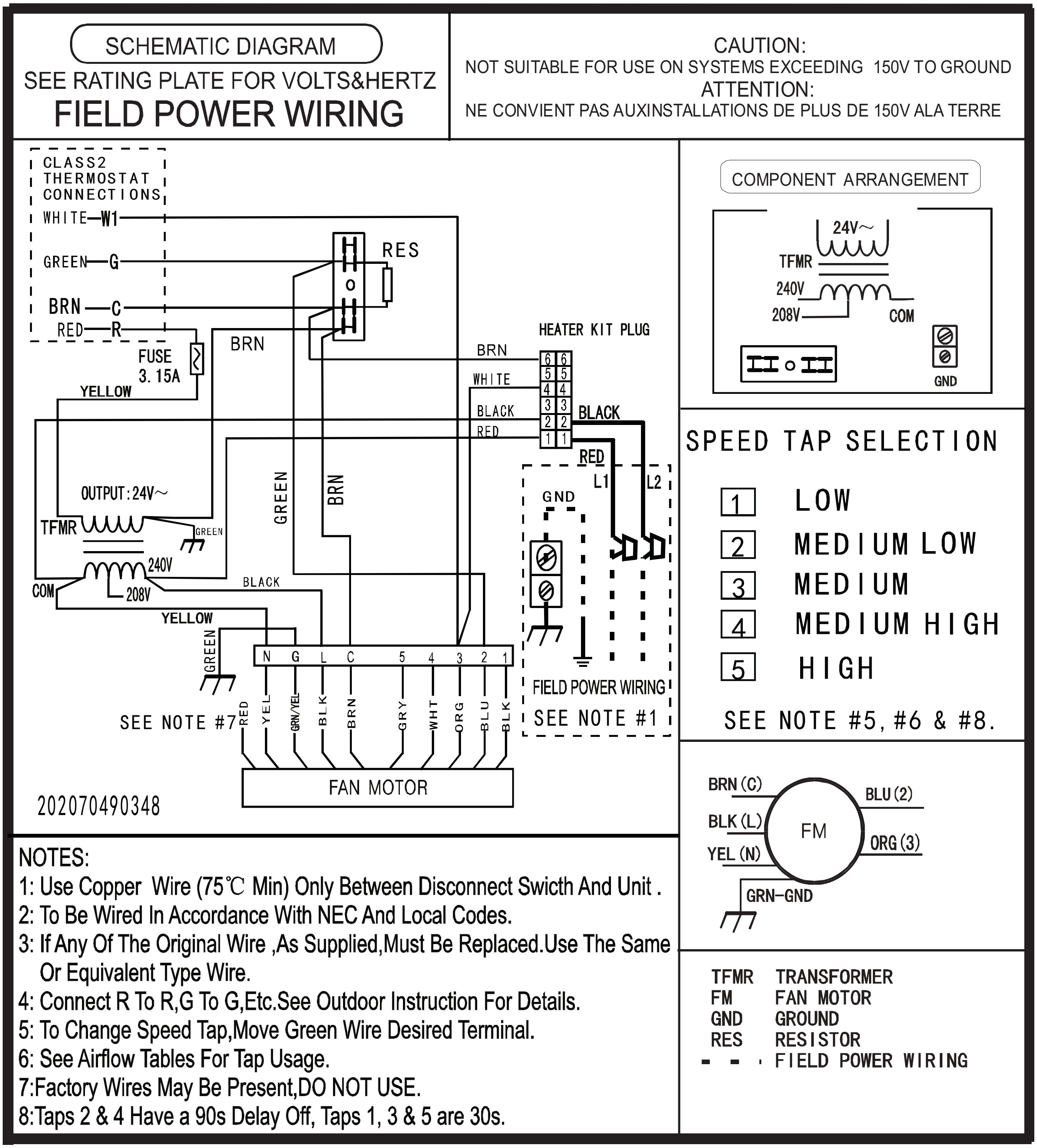
Fig. 23 – FFMANP(018,024,030,036 with PCB Time Delay

NOTE: Representative for FFMANP(018,024,030,036) serial number date code 1715V and later.



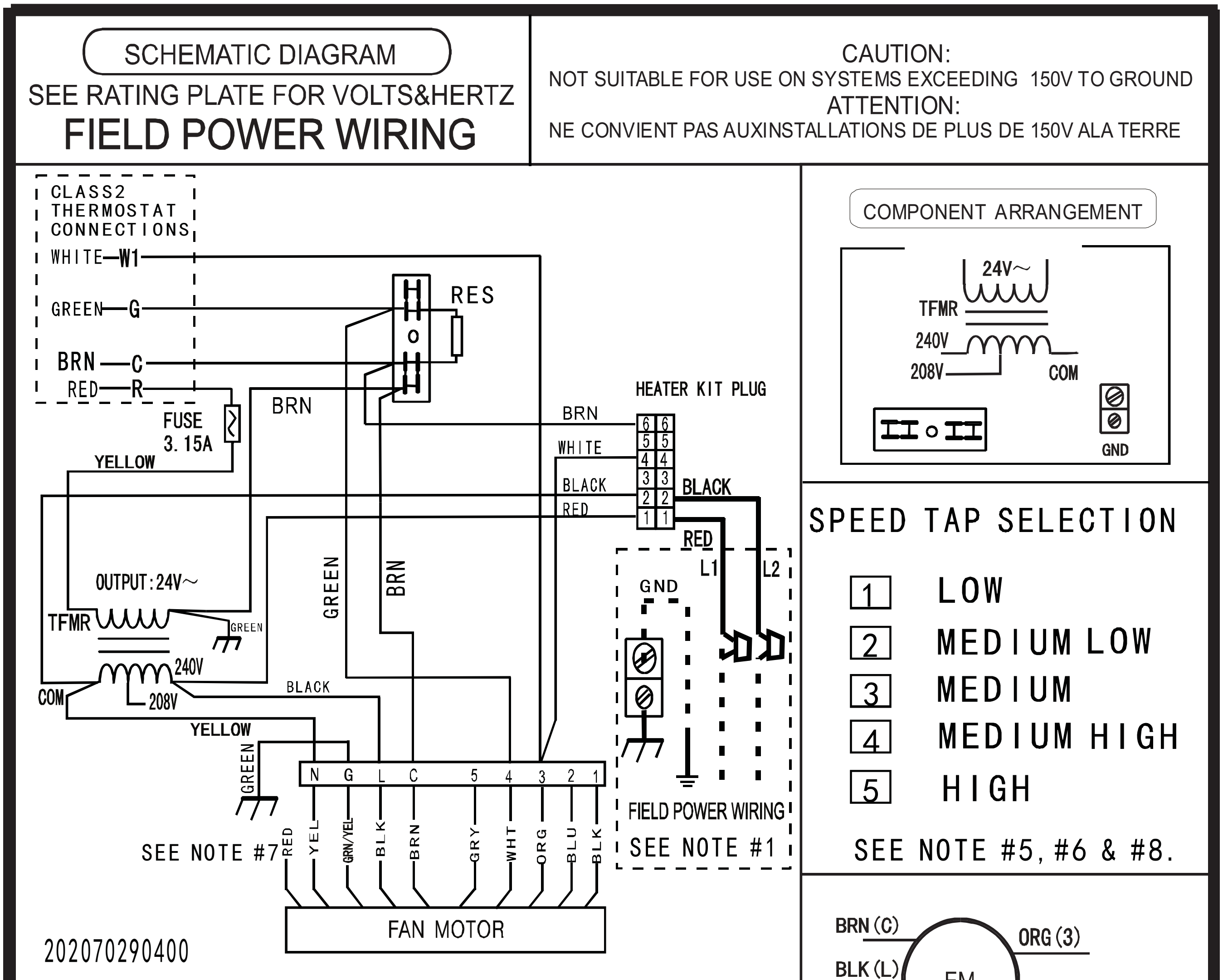
A150100

Fig. 24 - EHK2 With Heater Relays



A14323

Fig. 25 - FFMANP(019, 031)



NOTES:
1: Use Copper Wire (75℃ Min) Only Between Disconnect Swicth And Unit .
2: To Be Wired In Accordance With NEC And Local Codes.
3: If Any Of The Original Wire ,As Supplied,Must Be Replaced.Use The Same Or Equivalent Type Wire.
4: Connect R To R,G To G,Etc.See Outdoor Instruction For Details.
5: To Change Speed Tap,Move Green Wire Desired Terminal.
6: See Airflow Tables For Tap Usage.
7:Factory Wires May Be Present,DO NOT USE.
8:Taps 2 & 4 Have a 90s Delay Off, Taps 1, 3 & 5 are 30s.

TFMR	TRANSFORMER
FM	FAN MOTOR
GND	GROUND
RES	RESISTOR
- - -	FIELD POWER WIRING

A14324

Fig. 26 - FFMANP(025, 037)

SCHEMATIC DIAGRAM

SEE RATING PLATE FOR VOLTS&HERTZ

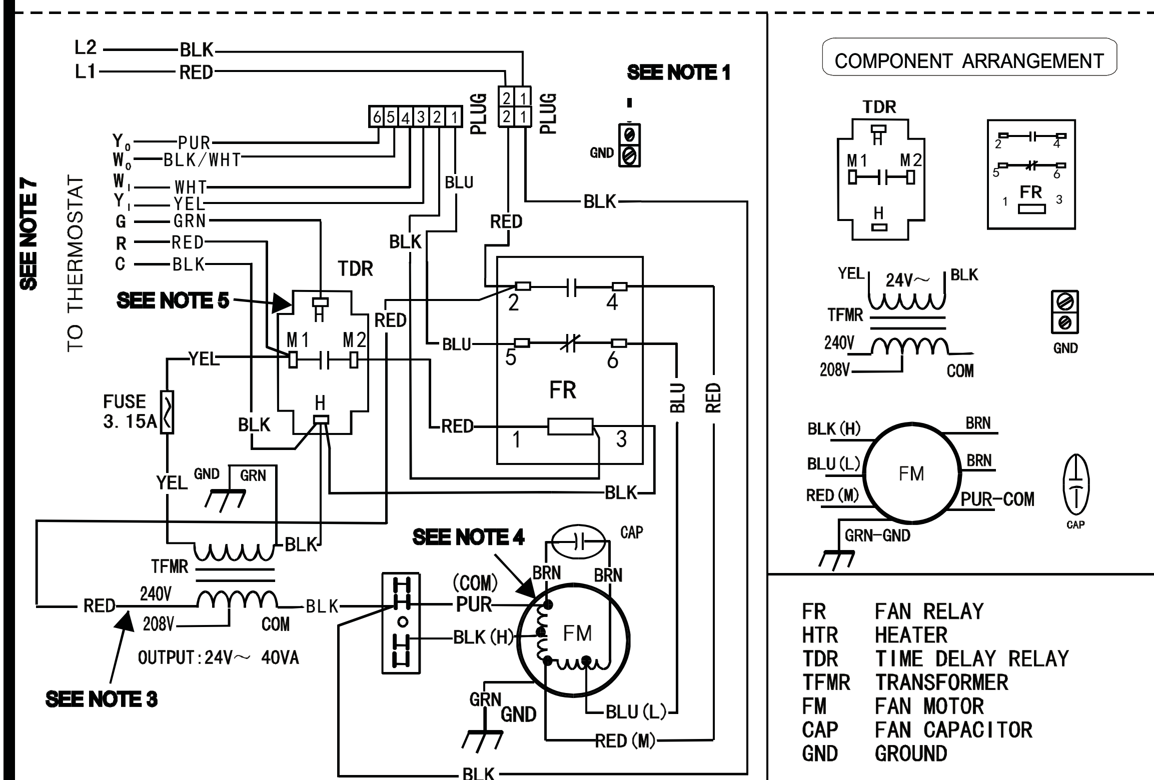
FIELD POWER WIRING

CAUTION:

NOT SUITABLE FOR USE ON SYSTEMS EXCEEDING 150V TO GROUND

ATTENTION:

NE CONVIENT PAS AUXINSTALLATIONS DE PLUS DE 150V ALA TERRE



NOTES:

1: Use copper wire(75℃ min) only between disconnect switch and unit, To be wired in accordance with N.E.C. and local codes. Fan coils equipped with electric heater connect power supply to terminal block.Cooling controls wiring not used with electric heaters.

2: If any of the original wire as supplied must be replaced,use the same or equivalent type wire.

3: Remove the red lead from "240V" terminal and then connect the red lead to "208V" terminal on the transformer for 208 volts.

4: Factory default fan speed is Medium,FM red wire connected to FR #4;For HI speed connect FM black wire to FR #4;For LOW speed connect FM blue wire to FR #4,and FM red wire connected to FR #6 . Always connect the unused FM wire to the dummy terminal block.

5: TDR has a 1-20s on delay when "G" is energized and a 50-70s off delay when "G" is de-energized.

6:Connect R to R, G to G, etc. See outdoor or indoor instructions for details.

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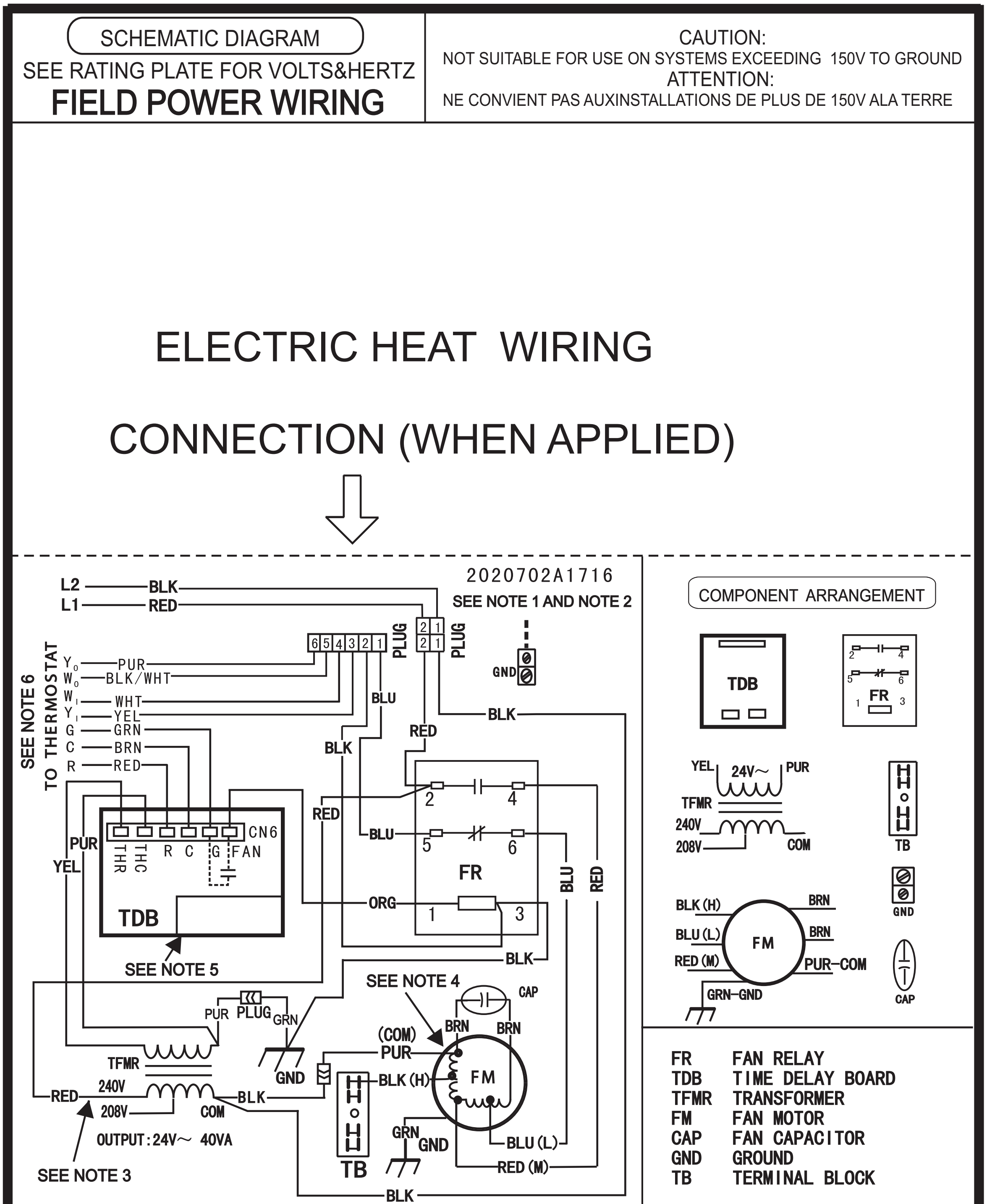
Fig. 27 - FPM(A,B)N(U,C) with Time Delay Relay



NOTES:

1: Use copper wire(75℃ min) only between disconnect switch and unit, To be wired in accordance with N.E.C. and local codes. Fan coils equipped with electric heater connect power supply to terminal block.Cooling controls wiring not used with electric heaters.

2: If any of the original wire as supplied must be replaced,use the same or equivalent type wire.

3: Remove the red lead from "240V" terminal and then connect the red lead to "208V" terminal on the transformer for 208 volts.

4: Factory default fan speed is Medium,FM red wire connected to FR #4;For HI speed connect FM black wire to FR #4;For LOW speed connect FM blue wire to FR #4,and FM red wire connected to FR #6 . Always connect the unused FM wire to the dummy terminal block.

5: TDB has a 90-100s off delay when "G" is de-energized.

6:Connect R to R, G to G, etc. See outdoor or indoor instructions for details.

7:N.E.C.Class 2,24volts.

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Fig. 28 – FPM(A,B)N(C,U) with Time Delay Board

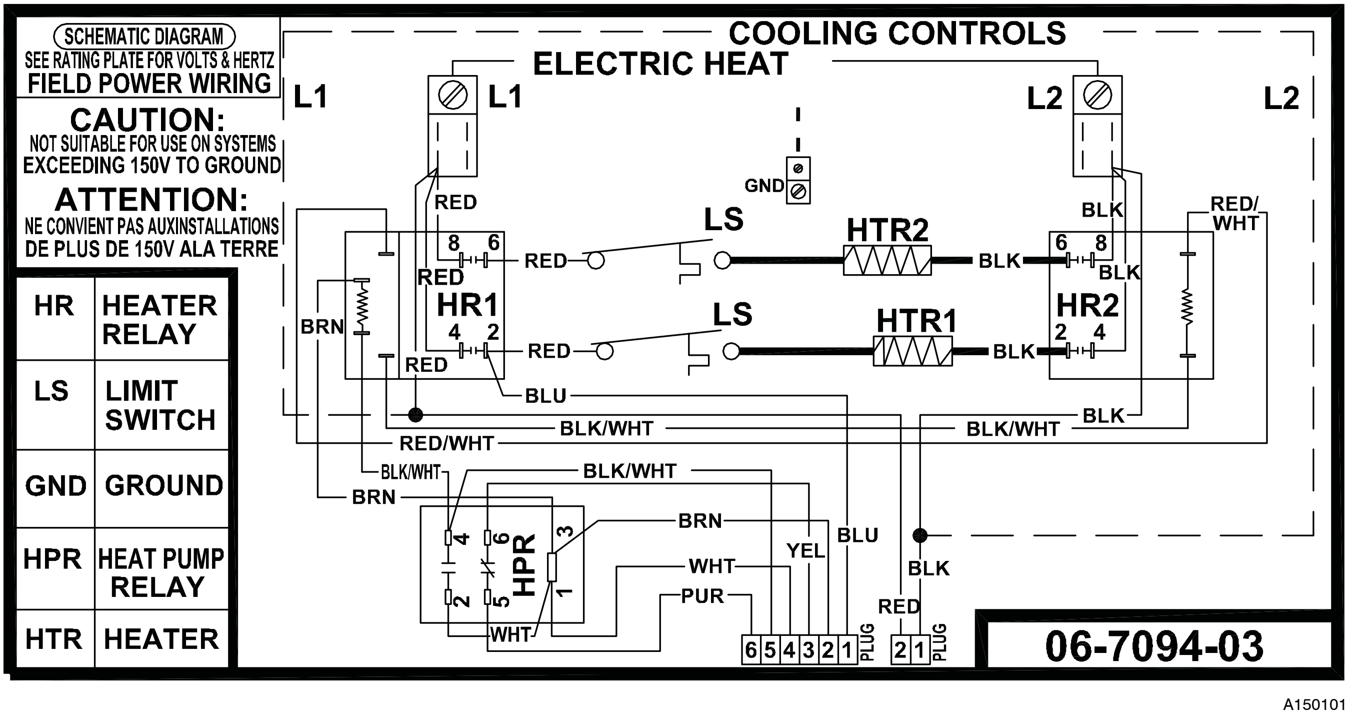
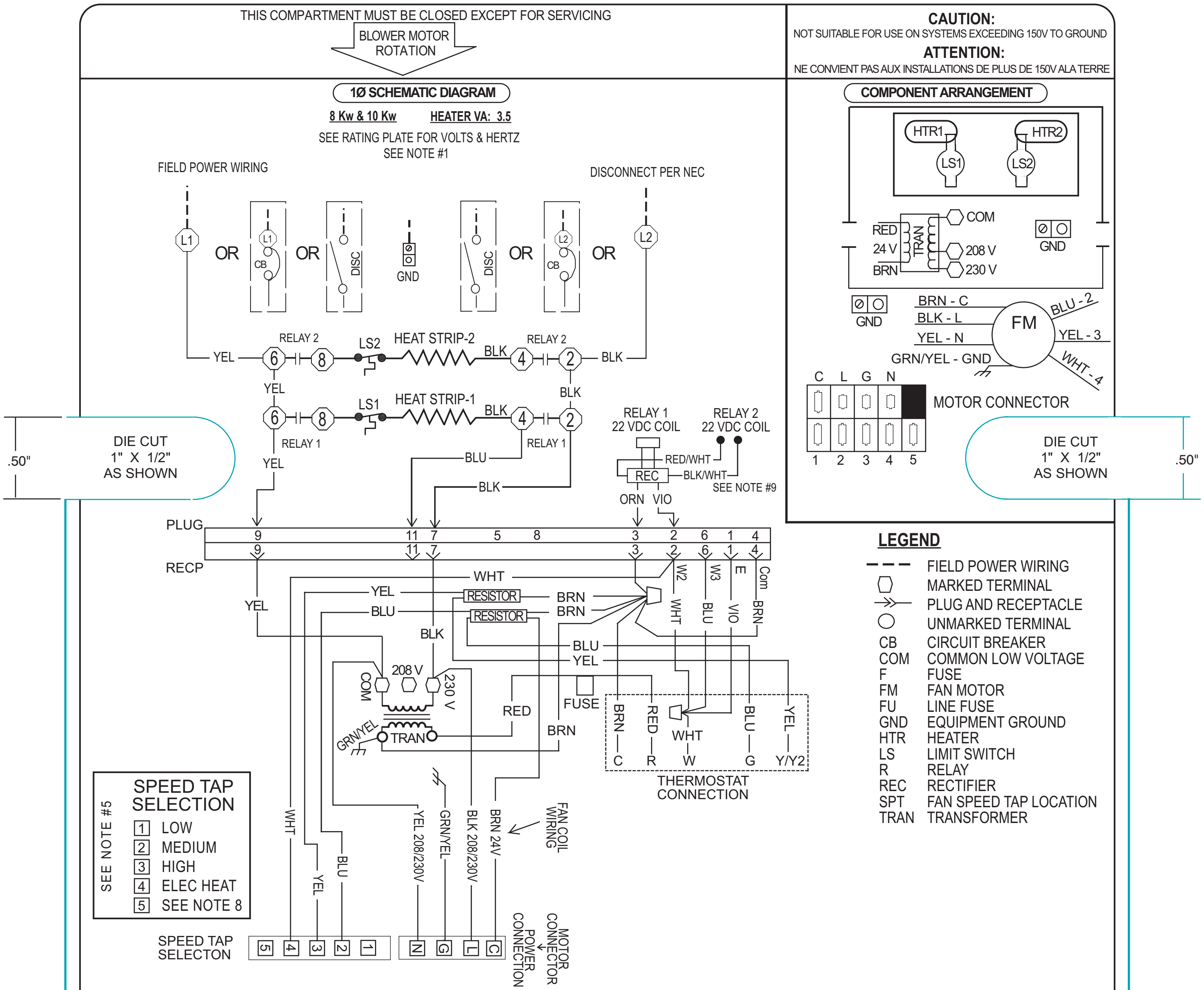


Fig. 29 - EHK3 With Heater Relays



NOTES:

1. Use copper wire (75°C min.) only between disconnect switch and unit.
2. To be wired in accordance with N.E.C. and local codes.
3. If any of the original wire, as supplied, must be replaced, use the same or equivalent type wire.
4. Replace low voltage fuse with no greater than 5 amp fuse.
5. To change speed tap, move blue wire to desired terminal.
6. Smaller heaters will have fewer components.
7. Connect R to R, G to G, etc., see outdoor instruction for details.
8. See airflow tables for tap usage.
9. Red Wire/WHT Stripe to QC1A, BLK Wire/WHT Stripe to QC2A.
10. N.E.C. class 2, 24 volts.

342416-101 REV. A

A160102

Fig. 30 - 342416-101

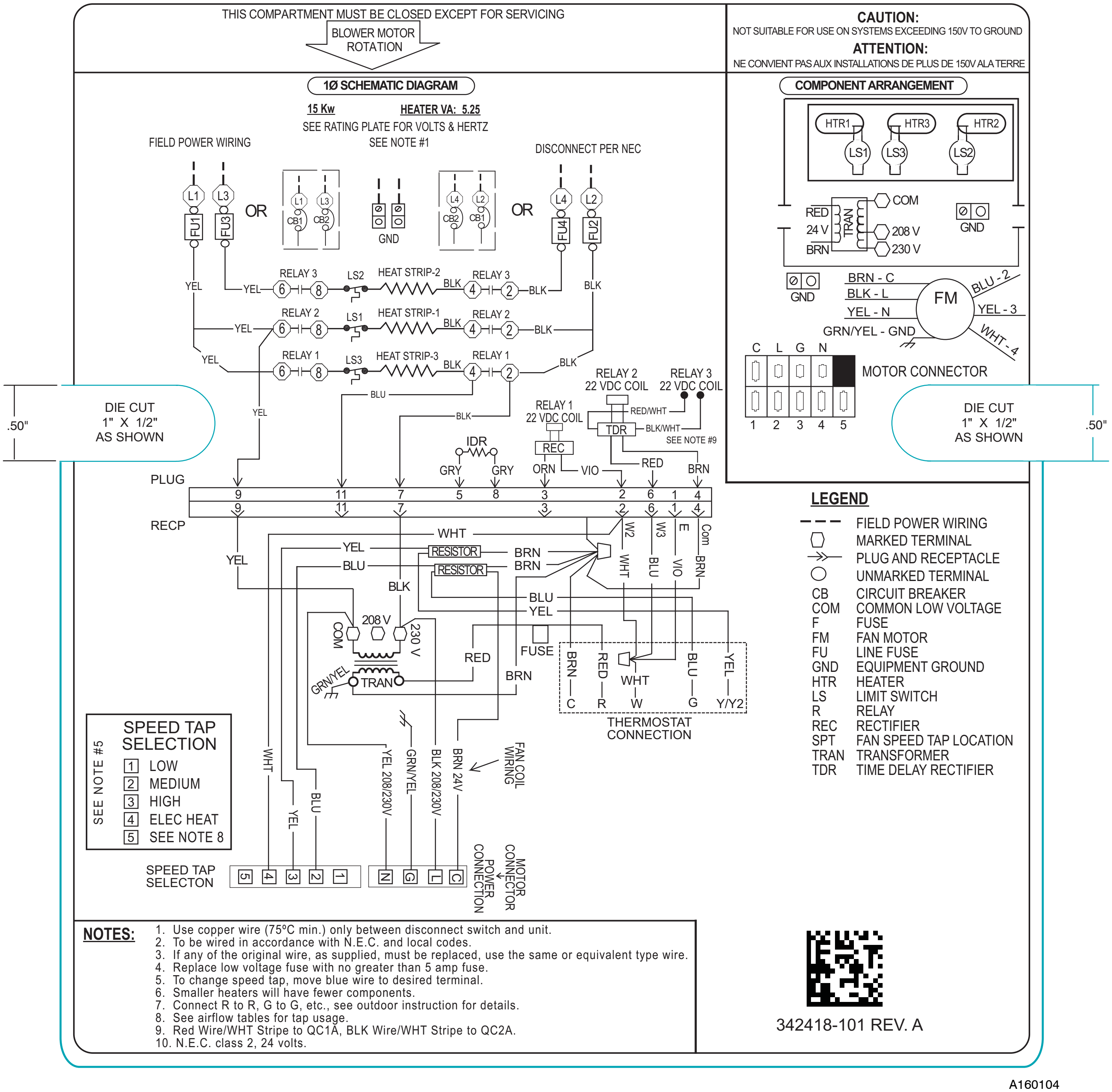
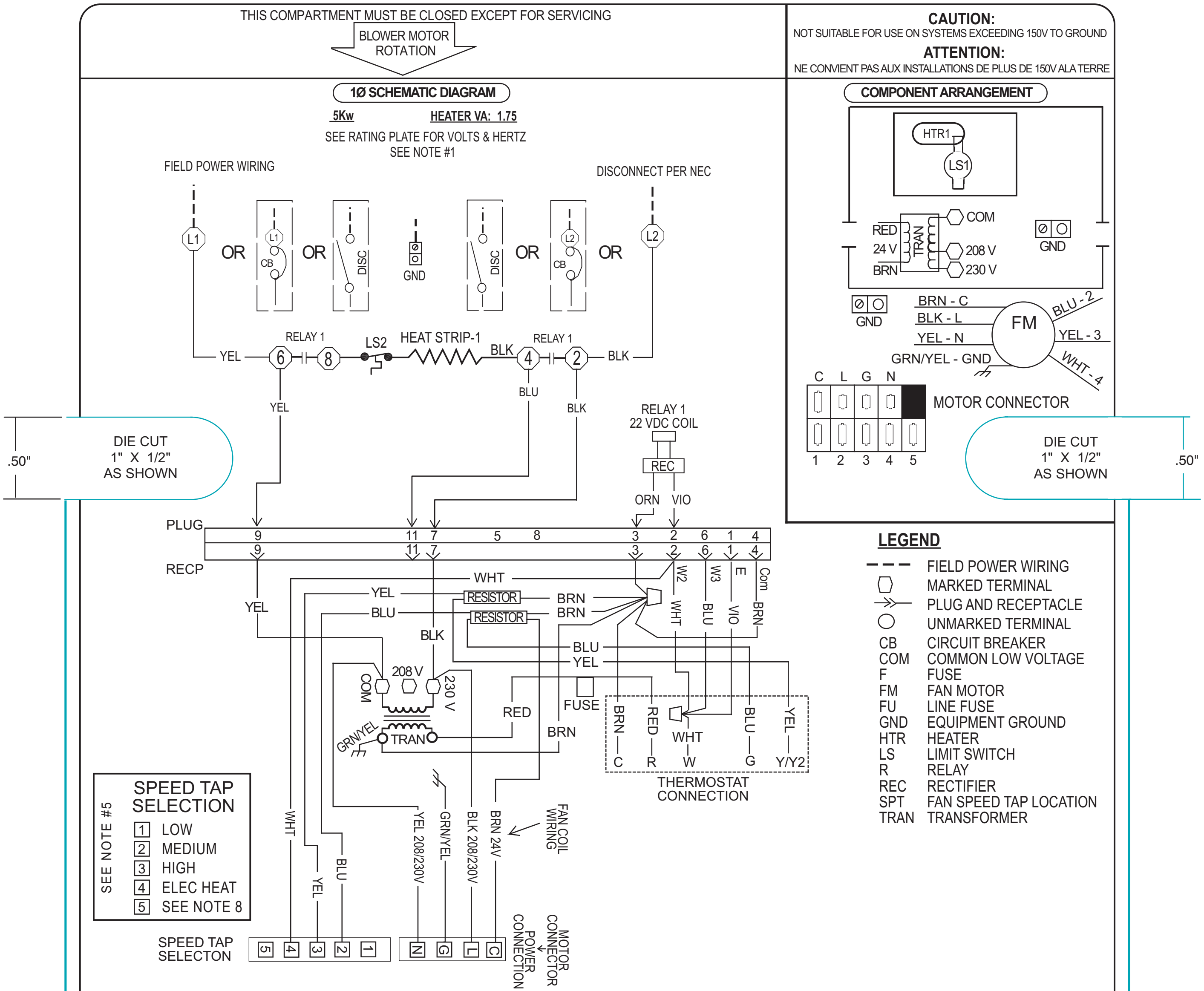


Fig. 31 - 342418-101



NOTES:

1. Use copper wire (75ºC min.) only between disconnect switch and unit.
2. To be wired in accordance with N.E.C. and local codes.
3. If any of the original wire, as supplied, must be replaced, use the same or equivalent type wire.
4. Replace low voltage fuse with no greater than 5 amp fuse.
5. To change speed tap, move blue wire to desired terminal.
6. Smaller heaters will have fewer components.
7. Connect R to R, G to G, etc., see outdoor instruction for details.
8. See airflow tables for tap usage.
9. N.E.C. class 2, 24 volts.

342417-101 REV. A

A160103

Fig. 32 - 342417-101

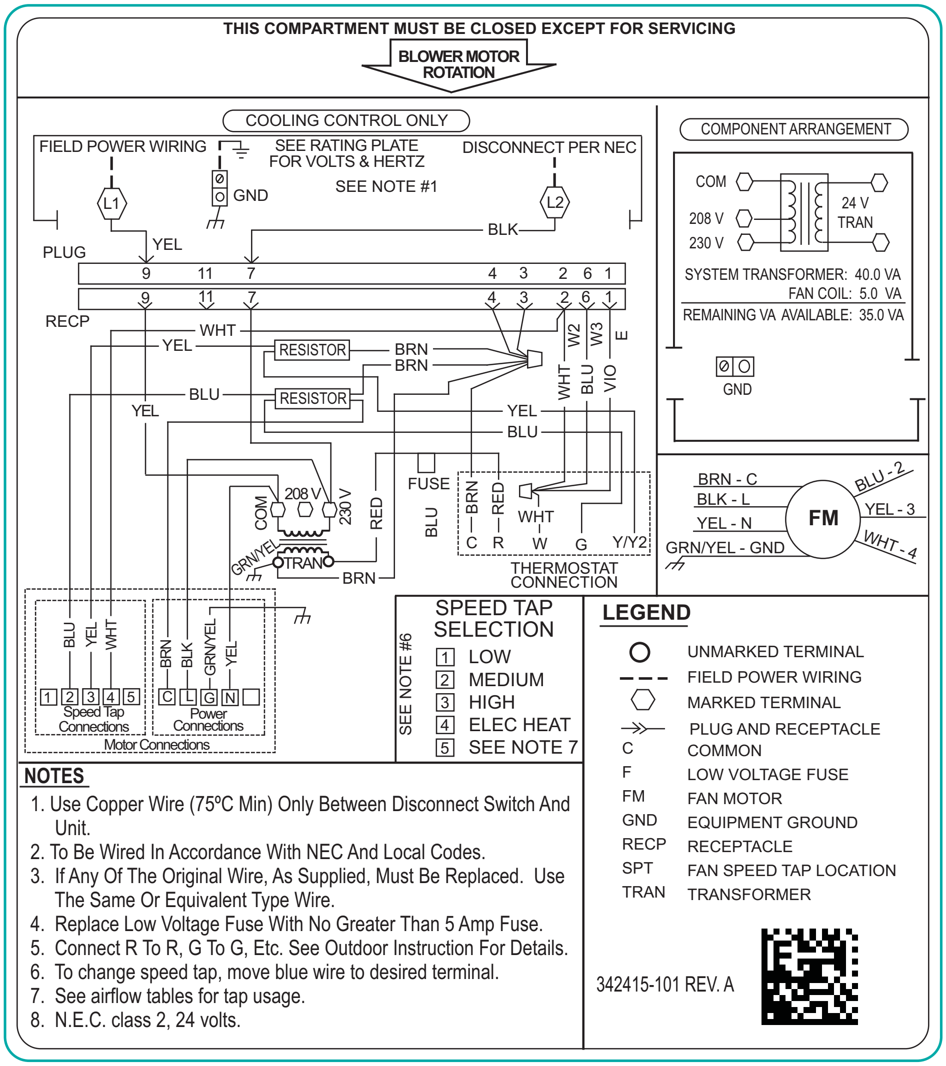


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Fig. 33 - FZ4A